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BACHELOR'S THESIS

Modelling Wellbeing: Comparing Statistical,
Machine Learning, Hybrid and Bayesian Network
Models on Psychological Data

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Modellierung von Wohlbefinden: Vergleich
statistischer, maschineller Lern-, hybrider und
bayesscher Netzwerkmodelle anhand
psychologischer Daten

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LUCRARE DE LICENȚĂ

Modelarea bunăstării: compararea modelelor
statistice, de învățare automată, hibride și de rețele
bayesiene pe date psihologice

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Abstract

Psychological wellbeing is shaped by a complex interplay of personal, social, economic, and environmental factors operating at both individual and societal levels. This thesis presents a systematic comparative study of statistical, machine learning, hybrid, and Bayesian Network models for predicting and explaining wellbeing outcomes across four datasets spanning two levels of analysis: workplace mental health, student depression, individual life satisfaction, and national happiness.

Predictive performance is driven primarily by data quality and signal concentration rather than algorithm choice. Ensemble methods consistently outperform simpler baselines, with macro-level aggregated data substantially more predictable than individual-level survey responses. Hybrid models fail to outperform well-configured baselines across all datasets. Unconstrained Bayesian Network structure learning produces high rates of causally implausible edges, confirming that data-driven causal discovery requires domain constraints to be reliable. Expert-constrained Bayesian Networks recover theoretically coherent causal structures at negligible statistical cost. SHAP analysis identifies proxy predictors whose removal improves causal interpretability while preserving predictive performance.

These findings demonstrate that machine learning and Bayesian Network models serve complementary rather than competing purposes: machine learning optimises predictive accuracy while Bayesian Networks recover interpretable causal structure, contributing empirical evidence to the prediction-explanation trade-off debate in psychological and wellbeing research.

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Chapter 1 - Introduction

1.1. Motivation

Mental health and psychological wellbeing are among the most complex and consequential outcomes in modern society. Conditions such as depression, chronic stress, and low life satisfaction affect hundreds of millions of people worldwide, with significant consequences for individuals, organisations, and public health systems [Helliwell2023]. Despite a broad scientific consensus on the factors that contribute to wellbeing - including social support, economic security, physical health, and psychological resilience [Diener1984; OECD2020] - the question of how accurately and interpretably these factors can be modelled using computational methods remains open [Breiman2001; Shmueli2010].

Two parallel challenges motivate this thesis. The first is the distinction between individual-level and societal-level data. Survey responses from employees or students capture rich personal experiences but are characterised by high noise, subjective bias, and complex interactions among predictors [Yarkoni2017; Shmueli2010]. Aggregate indicators such as GDP per capita, life expectancy, and institutional trust capture structural conditions at the country level but abstract away individual variation [Helliwell2023; OECD2020]. Whether predictive models perform differently across these two levels of analysis - and whether the same modeling frameworks can be applied consistently to both - is a question with both methodological and practical implications [Breiman2001].

The second challenge concerns the tension between prediction and explanation [Breiman2001; Shmueli2010]. Machine learning models have demonstrated strong predictive performance across a wide range of psychological and health outcomes [Hastie2009; Kuhn2013], but their internal logic is often opaque [Molnar2019]. Statistical models offer interpretability but may miss nonlinear relationships [Hastie2009]. Bayesian Networks occupy a distinctive position: they are generative probabilistic models that can encode causal structure explicitly, enabling reasoning under uncertainty in a way that neither regression nor ensemble methods can match [Pearl2009; Scutari2014]. Whether Bayesian Networks can recover plausible causal structures from observational wellbeing data - and whether those structures add value beyond what machine learning feature importance already reveals [Lundberg2017] - is the central methodological question of this thesis [Pearl2009, Scutari2014].

1.2. Scope

This thesis presents a systematic comparative study of statistical, machine learning, hybrid, and Bayesian Network models for predicting and explaining psychological wellbeing under uncertainty, applied consistently across both individual-level and societal-level data.

The study is grounded in four datasets spanning two levels of analysis. At the individual level, three datasets are examined: a survey of mental health conditions in the technology industry [OSMI2016], a survey of depression among students [Kaggle_Depression], and a Canadian population health survey measuring life satisfaction [Kaggle_CCHS]. At the societal level, the World Happiness Report Gallup Poll dataset - covering 166 countries across 2005–2020 - provides a macro-level comparison [Kaggle_WHR]. An additional synthetic Kaggle dataset nominally derived from the World Happiness Report is included as a data quality case study, demonstrating the consequences of absent signal for all modelling approaches [Kaggle_WHRSynthetic]. All datasets were obtained from Kaggle, an open-access platform for public datasets [Kaggle].

The scope is deliberately comparative. The same modeling frameworks are applied consistently across all datasets, enabling direct cross-dataset and cross-model evaluation. The study does not aim to perform clinical diagnosis or psychological assessment. Its focus is on modelling self-reported mental health and wellbeing outcomes using different predictive approaches, with particular emphasis on the causal structure of the data and the interpretability of model outputs.

Wellbeing is operationalised across multiple constructs depending on the dataset: work interference and treatment-seeking behaviour in the workplace context [Diener1984], depression as a binary clinical indicator in the student context, life satisfaction as a continuous self-report measure in the Canadian survey [Pavot1993], and the Cantril Life Ladder score as a societal happiness measure in the Gallup dataset [Helliwell2023, OECD2020].

1.3. Research Questions

The study is organised around five research questions that together address the comparative performance, interpretability, causal validity, and robustness of the modelling frameworks examined:

RQ1. How do different modeling approaches perform when predicting wellbeing at individual versus societal levels? This question examines whether predictive performance - measured by classification metrics at the individual level and regression metrics at the societal

level - differs systematically between data levels, and whether that difference varies across model families.

RQ2. How does interpretability differ between regression, machine learning, and Bayesian Networks? This question examines the qualitative and quantitative differences in how each model family explains its predictions, using coefficient analysis, SHAP values, feature importance rankings, and Bayesian Network conditional probability tables as complementary interpretability tools [Lundberg2017, Molnar2019, Goldstein2015].

RQ3. Are Bayesian Networks better at capturing plausible causal structures in wellbeing data? This question evaluates whether Bayesian Network structure learning, with and without expert constraints, recovers directed acyclic graphs whose edge directions are consistent with established domain knowledge about the causal determinants of mental health and wellbeing [Pearl2009, Morgan2015].

RQ4. Can Bayesian Networks reveal causal structures that ML models cannot? This question examines whether the explicit directional edges learned by Bayesian Networks provide causal information beyond what machine learning feature importance conveys, and what the conditions are under which data-driven structure learning succeeds or fails [Yarkoni2017, Scutari2014].

RQ5. How robust are these models to missing data, noise, and correlated predictors? This question evaluates model stability across cross-validation folds, sensitivity to the removal of dominant predictors, and performance under conditions of high missingness and correlated feature sets [Hastie2009, Kuhn2013].

1.4. Overview of Approach

The methodology follows a consistent pipeline applied to all datasets. Each dataset is preprocessed in two versions: a machine learning version with imputation, one-hot encoding, and missingness indicators, and a Bayesian Network version preserving missing values as discrete states for structure learning. Baseline machine learning models - logistic or linear regression, decision tree, random forest, and XGBoost - are trained and evaluated using stratified cross-validation with multiple metrics. Hybrid models combining nonlinear feature selection with linear estimators, and stacking ensembles, are evaluated against the baselines. Bayesian Networks are learned using Hill-Climb Search with BIC scoring, both unconstrained and with expert-defined forbidden edges encoding domain knowledge about causal ordering.

Interpretability is analysed at two levels. At the global level, SHAP TreeExplainer values provide model-agnostic feature importance and directional effect estimates for the best-performing model on each dataset. Sensitivity analyses remove dominant proxy predictors to assess robustness. At the structural level, Bayesian Network graphs are inspected for causally implausible edges, and BIC scores are compared between unconstrained and constrained versions to quantify the statistical cost of expert constraints.

Cross-dataset comparison is performed through unified performance tables, visualisations of hybrid model gain over baselines, and a Bayesian Network consolidation summarising structure quality, inference results, and causal pathway robustness across all datasets. The thesis concludes with a discussion connecting empirical findings to the five research questions and to the broader prediction-explanation debate in psychological science [Breiman2001, Shmueli2010].

Chapter 2 – Literature Review

2.1. Overview

This chapter situates the thesis within three intersecting bodies of literature: the debate between predictive and explanatory modelling in statistical and psychological science, the theory and practice of Bayesian Network causal discovery, and the measurement of subjective wellbeing at individual and societal levels. Together these three strands provide the conceptual foundation for the comparative modelling framework developed in this thesis and motivate the five research questions introduced in Chapter 1.

2.2. Prediction vs Explanation Debate

The relationship between statistical explanation and predictive accuracy has been a central tension in quantitative science for decades. The distinction was formalised most sharply by Breiman [Breiman2001], who identified two cultures in statistical modelling: the data modelling culture, which assumes a stochastic data-generating process and prioritises interpretable parametric models, and the algorithmic modelling culture, which treats the process as unknown and prioritises predictive accuracy. Breiman argued that psychology and the social sciences had become overly committed to the data modelling culture at the expense of predictive validity, and that algorithmic methods such as ensemble trees and neural networks offered substantial gains in out-of-sample performance.

Shmueli [Shmueli2010] developed this distinction more formally by introducing a framework for distinguishing explanatory modelling - aimed at testing causal hypotheses about the data-generating process - from predictive modelling - aimed at producing accurate forecasts of new observations. She argued that the two goals require different model selection criteria, different evaluation metrics, and different interpretive standards, and that conflating them leads to both poor predictions and poor causal inference. A model that fits the training data well and has theoretically grounded coefficients may nevertheless perform poorly on held-out data, while a model with strong out-of-sample performance may be causally uninterpretable. This framework directly motivates the comparative design of this thesis, which evaluates both predictive performance and causal interpretability across model families.

Yarkoni and Westfall [Yarkoni2017] extended this argument to psychological science, demonstrating that psychology's focus on causal explanation had produced a literature of intricate

theories with limited predictive validity, and proposing that machine learning principles - regularisation, cross-validation, and out-of-sample evaluation - would produce more generalisable psychological models.

The growing adoption of complex machine learning models in applied settings has generated a parallel literature on post-hoc interpretability. Hastie, Tibshirani, and Friedman [Hastie2009] provide the statistical foundation for ensemble methods and the conditions under which feature importance measures are reliable. Lundberg and Lee [Lundberg2017] unified feature importance methods under the SHAP framework, providing a theoretically grounded decomposition of predictions into additive feature contributions based on Shapley values, satisfying axiomatic properties that earlier importance measures do not. Molnar [Molnar2019] situates SHAP alongside other explanation methods within a comprehensive taxonomy of interpretable machine learning. In this thesis, SHAP is applied as the primary interpretability tool for the best-performing ML model on each dataset, complementing the structural interpretability provided by Bayesian Network graphs and coefficient tables from linear models.

2.3. Bayesian Networks Causal Discovery

Bayesian Networks are directed acyclic graphical models representing a joint probability distribution as a product of conditional distributions [Scutari2014]. Each node represents a variable, each directed edge a conditional dependency, and the absence of an edge encodes conditional independence - making Bayesian Networks a natural language for expressing causal hypotheses about data-generating processes.

Pearl [Pearl2009] established the theoretical foundations for causal inference in graphical models, introducing the do-calculus to distinguish interventional from observational quantities, and the d-separation criterion as a graphical test for conditional independence. A fundamental limitation of data-driven structure learning is that it cannot distinguish between Markov equivalent graphs - directed acyclic graphs encoding the same conditional independences but differing in edge direction. Without domain constraints, structure learning algorithms may recover statistically plausible but causally incoherent structures.

In this thesis, expert knowledge is incorporated by specifying forbidden edges encoding known causal ordering constraints - for example, that demographic variables cannot be caused by psychological outcomes. The technical implementation is described in Section 3.5; the impact of

these constraints on statistical fit and causal plausibility is evaluated systematically across all datasets.

2.4. Wellbeing Measurement

Subjective wellbeing is a multidimensional construct encompassing cognitive evaluations of life quality, positive affect, and negative affect [Diener1984]. Diener's foundational review identified three primary components - life satisfaction, positive affect, and negative affect - that have remained the dominant framework in wellbeing research for four decades and provide the theoretical grounding for the outcome variables used in this thesis.

At the individual level, life satisfaction is commonly measured using single-item or multi-item numeric scales. The Canadian Survey dataset uses a 0–10 single-item life satisfaction rating, consistent with established large-scale population survey instruments [Pavot1993]. The MH Tech and Student Depression datasets use self-reported interference with work and depressive symptom status respectively - binary and ordinal outcomes that capture individual-level psychological wellbeing in occupational and educational contexts.

At the societal level, the World Happiness Report provides the most comprehensive annual assessment of national happiness [Helliwell2023]. The primary measure is the Cantril Life Ladder - a 0–10 self-anchoring scale - with six explanatory factors: GDP per capita, social support, healthy life expectancy, freedom, generosity, and perceptions of corruption. These factors are precisely the predictors available in the Gallup dataset, enabling a direct empirical test of whether their importance rankings recovered by ML and BN models are consistent with the established WHR framework.

The OECD framework [OECD2020] motivates the inclusion of both objective socioeconomic variables - income, food security, employment - and subjective psychological states - stress, sense of belonging, mental health - as predictors in the Canadian Survey models.

Chapter 3 – Methodology

3.1. Overview

This chapter describes the methodological framework used to address the five research questions set out in Chapter 1. The study adopts a comparative design in which four families of models - statistical regression, machine learning, hybrid, and Bayesian Networks - are applied to five datasets varying in scale, measurement level, and task type. Two datasets involve classification of psychological outcomes at the individual level; one involves regression of life satisfaction at the individual level; and one involves regression of subjective wellbeing at the societal (macro) level. A fifth dataset, synthetically generated, is retained as a data quality case study rather than a substantive modelling exercise.

Section 3.2 describes each dataset and the preprocessing decisions applied to it. Section 3.3 introduces the four model families and their configurations. Section 3.4 covers the evaluation framework, including metrics, cross-validation strategy, and the handling of the evaluation asymmetry between ML and BN pipelines. Section 3.5 gives a full account of the Bayesian Network structure learning approach, covering the algorithm, scoring function, expert knowledge constraints, parameter estimation, and inference procedure.

3.2. Dataset descriptions and preprocessing decisions

Five datasets were selected to cover a range of wellbeing constructs, measurement instruments, and analytical challenges. The distinction between micro-level (individual survey data) and macro-level (country-aggregated data) is treated as a primary axis of comparison throughout the thesis.

3.2.1 Mental Health in Tech Survey

The OSMI Mental Health in Tech Survey is a publicly available dataset collected via self-report from technology-sector workers. After removing observations with missing target values, the working sample comprised 995 respondents and 43 features. The prediction target was a three-class variable, *combined_target*, constructed by crossing the work interference item (*work_interfere*) with whether the respondent had sought treatment: Class 0 indicates no interference, Class 1 indicates interference without treatment, and Class 2 indicates interference with treatment. The class distribution was approximately 60.6%, 21.4%, and 18.0% respectively.

Preprocessing for the machine learning pipeline included the addition of binary missingness indicators for features with non-trivial missing rates, followed by median imputation for continuous variables and mode imputation for categoricals. Nominal features were one-hot encoded with reference categories dropped to prevent multicollinearity. For the Bayesian Network pipeline, missing values were preserved as a distinct state rather than imputed [Scutari2014], ordinal encodings were retained, and high-cardinality geographic features (*Country* and *state*) were excluded.

3.2.2 Student Depression Dataset

This dataset contains survey responses from students on academic, social, and psychological variables. After filtering to confirmed student respondents - removing 0.1% of rows identified as non-students - the sample comprised 27,870 observations. The binary target, *Depression_yes*, indicated presence or absence of depression. Four features were dropped prior to modelling: *City*, which contained data entry inconsistencies; *Profession*, redundant after student filtering; and *Work Pressure* and *Job Satisfaction*, which are not applicable to a student population. The final feature set comprised 23 variables.

A notable characteristic of this dataset is the dominance of the suicidal ideation feature, which accounts for approximately 52–56% of intrinsic variable importance in tree-based models. Because suicidal ideation is a co-symptom of depression rather than an antecedent cause, its concentration of predictive signal raises questions about whether the models are learning genuine risk-factor structure or simply detecting a correlated symptom. This is examined in sensitivity analysis. For the Bayesian Network, age and CGPA were discretised into bins to satisfy the discrete variable requirement; continuous features in the ML pipeline were retained in their original form.

3.2.3 Canadian Community Health Survey

This large-scale national health survey comprises approximately 108,000 respondents. The continuous target, *Life_satisfaction*, is measured on a 0–10 scale. Columns with more than 60% missing values were dropped prior to modelling. After preprocessing, approximately 58 features were retained.

Two variables - *Mental_health_state* and *Gen_health_state* - are self-rated health variables that function as proxies for life satisfaction: they correlate strongly with the target but arguably reflect

a similar evaluative process rather than being independent causal inputs. Their contribution is examined in a dedicated sensitivity analysis.

3.2.4 WHR Synthetic Dataset

This dataset, presented on Kaggle as a World Happiness Report synthetic dataset, contains 4,000 observations across 10 countries and 20 years. The *Happiness_Score* column was dropped prior to modelling due to direct target leakage. Initial modelling revealed that all model families returned R^2 values near zero, and the maximum absolute correlation between any feature and the target was 0.049. It is retained as a case study in data quality detection, illustrating how a consistent modelling pipeline can identify a null-signal dataset before interpretability or causal analysis is attempted. The methodological implications of this finding are discussed in Section 4.4.1.

3.2.5 WHR Gallup Dataset

The Gallup World Poll data underlying the World Happiness Report was obtained via Kaggle and represents the most credible macro-level dataset in this study. The working sample comprised 1,949 observations across 166 countries and the period 2005–2020. The target variable, *Life_Ladder*, is the Cantril self-anchoring scale, which asks respondents to place their current life on a ladder from 0 (worst possible life) to 10 (best possible life). Seventeen features were retained after preprocessing, including log GDP per capita, social support, healthy life expectancy at birth, freedom to make life choices, generosity, perceptions of corruption, positive affect, negative affect, year, and binary missingness indicators for all columns with missing values. The maximum absolute feature-target correlation was 0.783, confirming genuine predictive signal. For the Bayesian Network, all continuous variables were quantile-discretised into five equal-frequency bins; year was binned into five-year periods to reduce the number of discrete states while preserving temporal ordering.

3.3. Model families and configurations

Four model families are applied consistently across all datasets. Each family represents a distinct position on the trade-off between predictive performance and interpretability, and their comparison is central to Research Questions 1, 2, and 3.

3.3.1 Statistical Baseline Models

The statistical baseline models serve as the interpretability anchor for the study. For classification tasks, logistic regression is used; for regression tasks, ordinary least squares linear regression is used. These models are selected not because they are expected to achieve the highest predictive accuracy, but because they are maximally transparent: every predictor's effect is directly readable from its estimated coefficient, including direction and magnitude [Shmueli2010]. This transparency makes them the natural reference point for the interpretability comparisons developed in later chapters.

A single decision tree (classifier or regressor depending on task) is also included in the baseline set. Decision trees occupy an intermediate position: they are non-parametric and capable of capturing non-linear relationships and interactions, yet their structure can be visualised and traversed directly. Maximum depth is constrained to limit overfitting [Hastie2009].

3.3.2 Machine Learning Models

Two ensemble methods form the primary machine learning models. Random Forest constructs a large ensemble of decision trees, each trained on a bootstrap sample of the training data with a random subset of features considered at each split [Breiman2001RF]. Predictions are aggregated by majority vote (classification) or averaging (regression). The randomisation introduced at both the sample and feature level reduces variance relative to any individual tree, and the ensemble's feature importance scores - based on the aggregate reduction in node impurity across all trees - provide a global measure of variable relevance.

XGBoost implements gradient boosting, constructing trees sequentially such that each new tree is fitted to the pseudo-residuals of the current ensemble under a differentiable loss function. Regularisation is applied through a learning rate (shrinkage), subsampling of rows and columns, and L1/L2 penalties on leaf weights [Chen2016]. XGBoost achieves the highest or near-highest predictive performance across all datasets in this study. Its predictions are used as the basis for the SHAP interpretability analysis described in Section 3.4.

Both models are evaluated using five-fold cross-validation with stratification on the target variable. Hyperparameters are held at well-established defaults throughout; the primary goal is comparison across model families rather than maximum optimisation of any individual model.

3.3.3 Hybrid Models

Hybrid models combine the feature selection capacity of ensemble methods with the interpretability of linear models. Two selection-then-estimation pipelines are evaluated: one using Random Forest importance scores to select features, followed by logistic or linear regression on the reduced feature set; and one using XGBoost importance scores in the same role. A third hybrid uses stacking, training a logistic or linear regression meta-learner on the out-of-fold predictions generated by Random Forest and XGBoost as base learners [Wolpert1992].

The inclusion of hybrid models reflects a practically motivated question: whether the interpretability of a linear model can be recovered without substantially sacrificing the predictive power of ensemble methods [Breiman2001].

3.4. Evaluation framework

Evaluation metrics are chosen to match the task type and to reflect the priorities established in the research questions.

For classification tasks - MH Tech and Student Depression - the primary metric is macro-averaged F1 score, which computes precision and recall separately for each class before averaging, weighting all classes equally regardless of their frequency in the data. This is appropriate given the class imbalance present in both datasets, where a majority-class classifier would score misleadingly well on accuracy. Area under the ROC curve (AUC) is reported as a threshold-independent complement; for the three-class MH Tech target, AUC is computed using a one-vs-rest strategy [Hastie2009].

For regression tasks - Canadian Survey, WHR Gallup, and WHR Synthetic - the primary metric is R^2 , the proportion of variance in the target explained by the model. Root mean squared error (RMSE) is reported alongside R^2 to provide a metric expressed in the original scale of the target variable, which aids interpretability of effect magnitudes.

All machine learning models are evaluated using stratified five-fold cross-validation, and the reported metrics are the mean values across the five held-out folds. Cross-validated metrics are the primary basis for performance comparisons, as they provide an estimate of out-of-sample generalisation rather than in-sample fit [Hastie2009]. Bayesian Network predictions are computed on the full dataset because the pgmpy inference pipeline does not readily support fold-level re-

estimation of structure and parameters. This evaluation asymmetry - CV metrics for ML models versus full-data metrics for BNs - is acknowledged throughout the results and discussion chapters.

3.5. BN structure learning approach

Bayesian Networks are probabilistic graphical models representing the joint distribution of a set of variables as a directed acyclic graph (DAG) [Pearl2009, Scutari2014]. As illustrated in Figure 3.1, each node represents a variable and each directed edge encodes a direct conditional dependence; each node's distribution is specified as a conditional probability table (CPT) over its parent states, and the full joint distribution factorises as the product of these local distributions - making inference tractable and the learned structure directly interpretable as a set of conditional independence claims. In Figure 3.1, the values are only illustrative, and variable names are drawn from the Student Depression dataset (Section 4.2) for concreteness.

All Bayesian Networks in this study are implemented using pgmpy version 1.1.0.

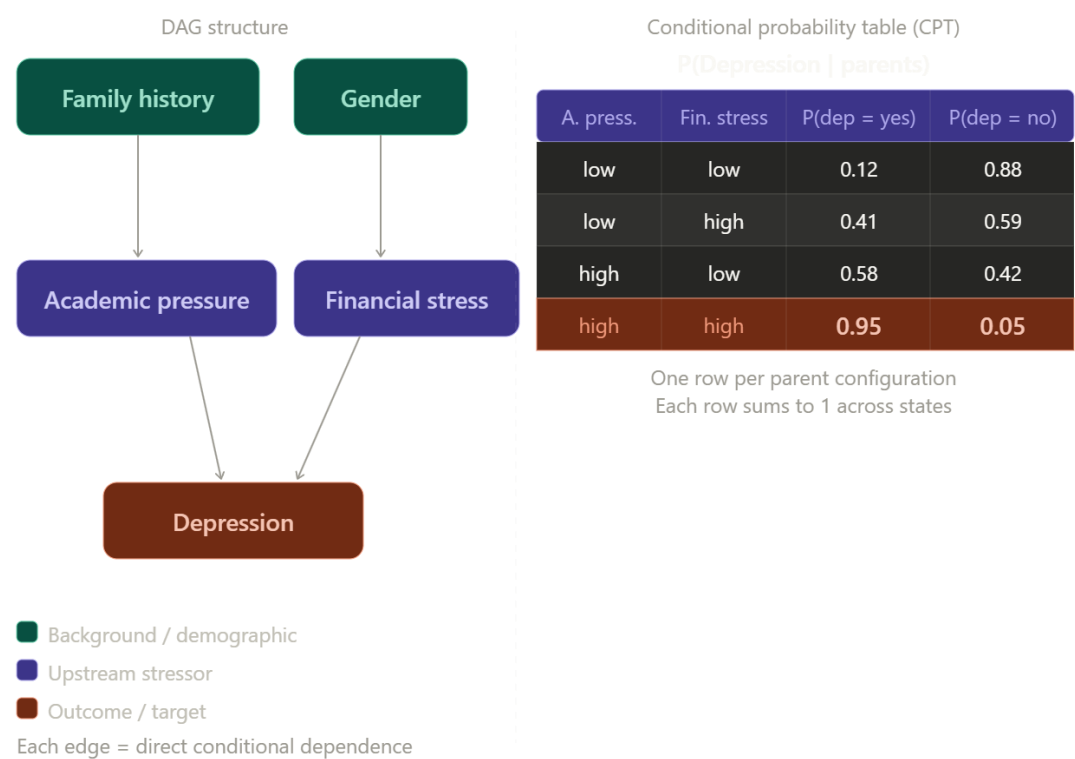


Figure 3.1. Hypothetical Bayesian Network illustrating DAG structure and conditional probability table

3.5.1 Structure Learning Algorithm

Structure learning is performed using the Hill Climb Search (HCS) algorithm, illustrated in Figure 3.2. As shown, HCS begins with an empty graph and iteratively proposes single-edge operations - addition, removal, or reversal - accepting whichever operation yields the greatest BIC improvement at each step and terminating when no improvement is found. HCS is a greedy search and does not guarantee a globally optimal solution, but is computationally practical for the variable counts in this study [Scutari2014].

To prevent overfitting through excessive parent sets, the maximum in-degree is constrained to three. This limits the combinatorial search space and produces structures that are more readily interpretable.

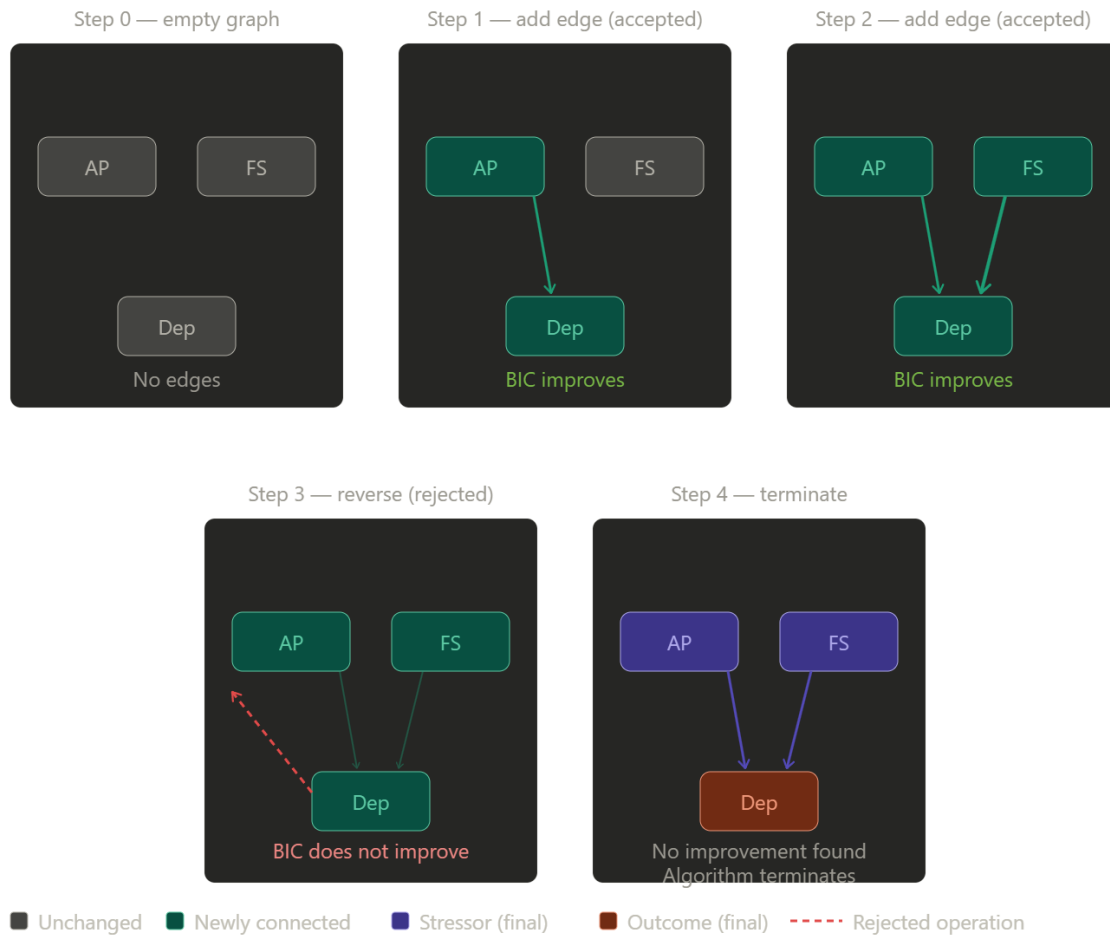


Figure 3.2. Hill Climb Search process illustrated on a three-node hypothetical graph. AP = Academic Pressure, FS = Financial Stress, Dep = Depression.

3.5.2 Scoring Function

The scoring function used throughout is BIC-d, the Bayesian Information Criterion adapted for discrete data. For a DAG G and dataset D of n observations:

$$\text{BIC}(G, D) = \log P(D \mid G, \theta) - (k / 2) \cdot \log n$$

where $\log P(D \mid G, \theta)$ is the log-likelihood under maximum likelihood parameter estimates and k is the total number of free parameters across all CPTs. As shown in Figure 3.3, HCS maximises this score - adding edges only when their likelihood gain outweighs the complexity penalty [Scutari2014].

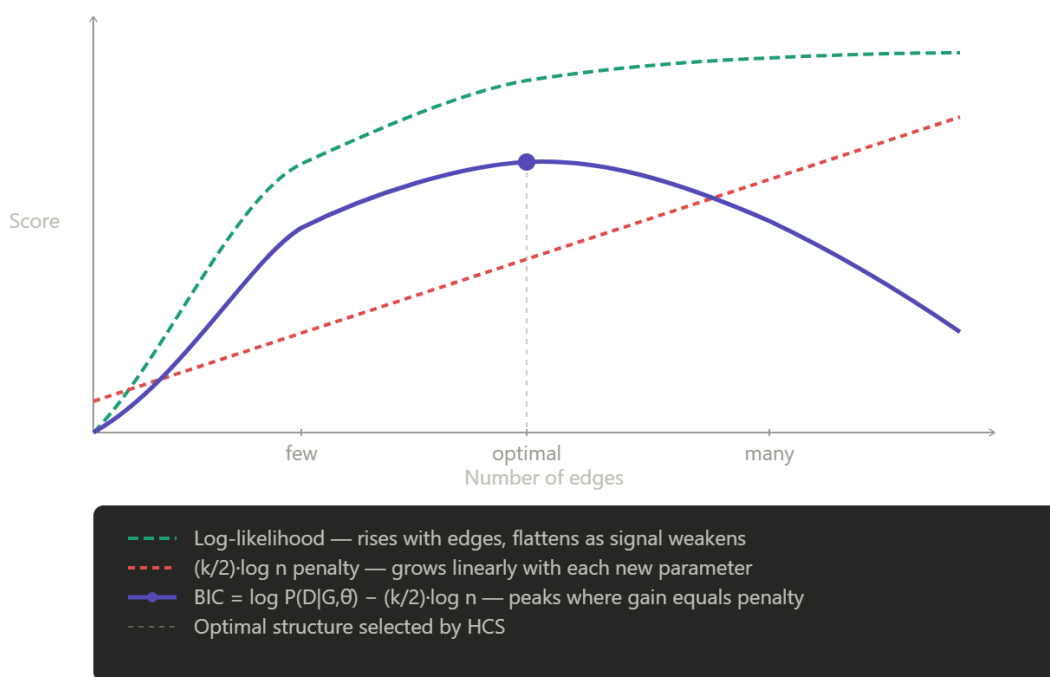


Figure 3.3. BIC scoring intuition: log-likelihood rises with edges while the penalty grows linearly; the BIC score peaks at the optimal structure.

3.5.3 Expert Knowledge Constraints

An unconstrained Hill Climb Search will sometimes recover edges that are statistically supported but domain-implausible - for example, Depression appearing as a parent of Academic Pressure,

reversing the theoretically motivated direction visible in Figure 3.1. To quantify this, an unconstrained BN is estimated for each dataset and all recovered edges are evaluated against domain knowledge, with the implausibility rate reported.

A restricted BN is then estimated by encoding domain constraints as a forbidden edges list, passed to HCS via pgmpy's ExpertKnowledge interface [Scutari2014]. Forbidden edge categories include: demographic or background variables as targets of outcome variables; time-invariant variables as targets of time-varying ones; and any edge whose direction is biologically, temporally, or theoretically impossible given the survey design.

For the Canadian Survey, an additional strongly restricted BN is estimated with `Mental_health_state` and `Gen_health_state` removed entirely, producing a three-way comparison: unconstrained, restricted, and strongly restricted.

The BIC cost of applying constraints is defined as:

$$\text{BIC cost (\%)} = ((\text{BIC_restricted} - \text{BIC_unconstrained}) / |\text{BIC_unconstrained}|) \times 100$$

A positive value indicates the restricted model achieves a better BIC score; a negative value indicates a fitness cost. Reporting this metric across datasets provides a quantitative characterisation of the tension between statistical fit and domain plausibility on each dataset.

3.5.4 Parameter Estimation

Conditional probability tables are estimated using the BayesianEstimator with a BDeu (Bayesian Dirichlet equivalent uniform) prior [Scutari2014]. BDeu assigns a uniform prior over the parameters of each CPT, distributed equally across all parent configurations, with the total prior weight controlled by the equivalent sample size hyperparameter. An equivalent sample size of 10 is used throughout, which provides mild smoothing of sparse cells without substantially distorting CPTs that are well-supported by data.

3.5.5 Inference and Prediction

Probabilistic inference is performed using Variable Elimination (VE). VE is an exact inference algorithm that computes the posterior distribution of a query variable given observed evidence by systematically eliminating non-query, non-evidence variables through marginalisation. As

illustrated in Figure 3.4, for a three-node chain $A \rightarrow B \rightarrow C$ with A observed, VE eliminates the intermediate variable B by summing over its states - $P(C | A=a) = \sum_b P(C | B=b) \cdot P(B=b | A=a)$ - returning the full conditional distribution over C without requiring enumeration of the joint distribution. In this study, VE returns $P(\text{target} | e)$ from which the most probable state (classification) or expected value from bin midpoints (regression) is used as the point prediction.



Figure 3.4. Variable Elimination on a three-node chain: A is observed, B is eliminated by marginalisation, and $P(C | A=a)$ is returned as the query result.

For regression targets quantile-discretised into bins, the expected value prediction is:

$$\hat{E}[\text{target} | e] = \sum_i \text{midpoint}_i \cdot P(\text{target} = \text{bin}_i | e)$$

where midpoint_i is the midpoint of the continuous range corresponding to bin i .

Chapter 4 - Results

4.1. Mental Health in Tech

4.1.1 Baseline ML Results

The four baseline models were evaluated on the three-class *combined_target* using five-fold stratified cross-validation. Results are reported in Table 4.1.

Table 4.1. MH Tech - Baseline ML cross-validated performance.

Model	CV Macro F1	CV AUC
Logistic Regression	0.457	0.685
Decision Tree	0.408	0.640
Random Forest	0.470	0.696
XGBoost	0.441	0.691

Random Forest achieved the highest macro F1 (0.470) and AUC (0.696) among the baseline models, with logistic regression and XGBoost performing comparably. The decision tree was the weakest baseline on both metrics. The narrow gap between logistic regression and Random Forest (0.013 F1 points) suggests the predictive structure is largely linear, with limited additional signal recoverable through non-linear modelling.

The overall performance level is modest, reflecting the difficulty of the three-class prediction task, class imbalance between the majority no-interference class (60.6%) and the two interference classes, and the inherent noise in self-report survey data.

As shown in Figure 4.1, the model performs strongest on Class 2 (interference with treatment), correctly classifying 96 of 120 instances, and weakest on Class 1 (interference without treatment), with only 10 of 36 correctly classified. This asymmetry suggests the intermediate class shares characteristics with both adjacent classes, making it the hardest to distinguish.

Figure 4.2 shows permutation feature importance for Random Forest. *family_history* dominates by a substantial margin, approximately three times larger than the next most important feature. Employer-level features - *care_options*, *supervisor*, *benefits* - appear in the mid-importance range, suggesting workplace environment contributes modestly beyond individual background factors.

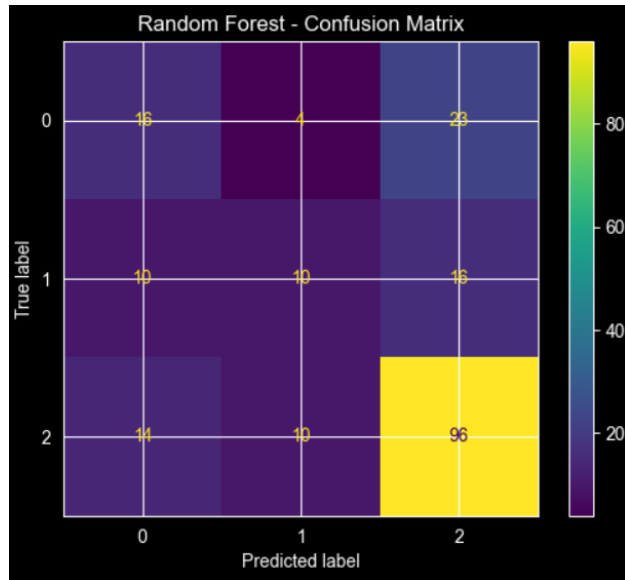


Figure 4.1 - Confusion Matrix: Best baseline model (Random Forest) for MH Tech Dataset

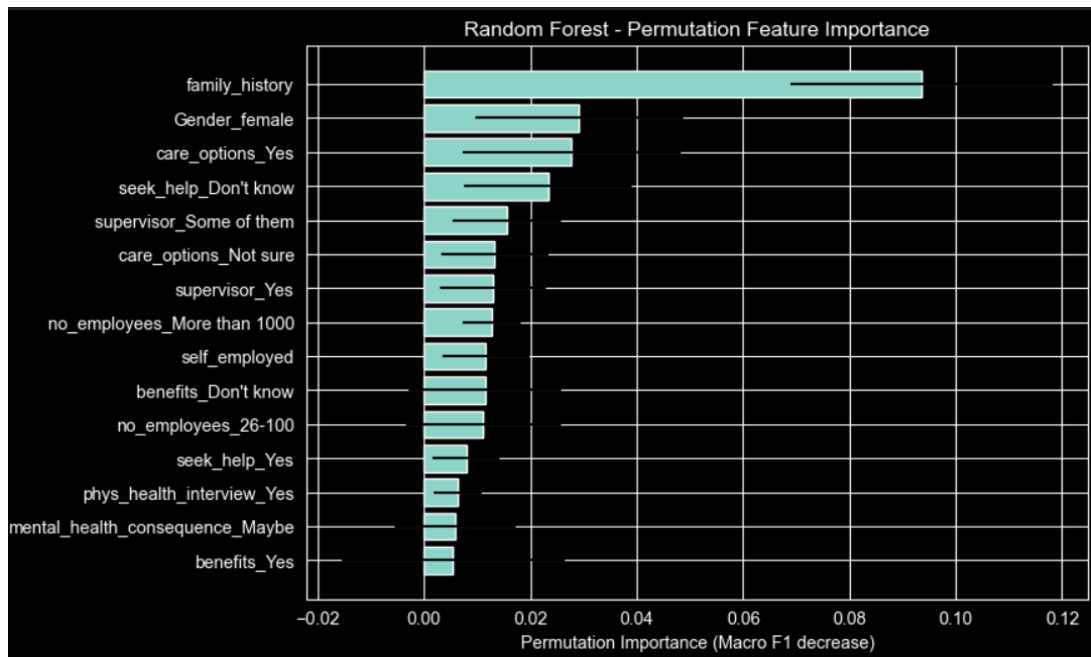


Figure 4.2 - Random Forest Permutation Feature Importance for MH Tech Dataset

4.1.2 Hybrid ML Results

Three hybrid configurations were evaluated: Random Forest feature selection followed by logistic regression (RF → LogReg), XGBoost feature selection followed by logistic regression (XGB → LogReg), and stacking with a logistic regression meta-learner. Results are reported in Table 4.2.

Table 4.2. MH Tech - Hybrid ML cross-validated performance

Model	CV Macro F1	CV AUC
RF Selection → LogReg	0.465	0.701
XGB Selection → LogReg	0.488	0.701
Stacking (LogReg meta)	0.460	0.700

The XGB → LogReg pipeline achieved the best hybrid F1 (0.488), a gain of 0.018 F1 points over the best baseline - the largest hybrid improvement observed across any dataset in this study yet still negligible in absolute terms. The cross-dataset pattern of hybrid model gains is discussed in Chapter 5.

4.1.3 BN Results

Two Bayesian Network versions were estimated: an unconstrained BN and a restricted BN incorporating domain-knowledge forbidden edges.

The unconstrained BN learned a graph containing 41 edges. Manual evaluation identified 10 as implausible, yielding an implausibility rate of 24%. The restricted BN was estimated after encoding a forbidden edges list to correct these structures, at a BIC cost of −3.98%.

The restricted BN produced the following direct parents for the target nodes:

- Parents of work_interfere: Gender, family_history
- Parents of treatment: Gender, work_interfere

This structure encodes a two-stage pathway in which gender and family history jointly influence the degree to which mental health affects work, and in which both gender and work interference influence whether an individual seeks treatment. Table 4.3 shows the full CPT for work_interfere under the restricted BN.

Table 4.3. CPT for work_interfere (restricted BN) - P(work_interfere | Gender, family_history).

Gender	Family History	P(Never)	P(Often)	P(Rarely)	P(Sometimes)
Female	No	0.159	0.148	0.183	0.510
Female	Yes	0.093	0.158	0.230	0.519
Male	No	0.358	0.111	0.172	0.360
Male	Yes	0.083	0.180	0.160	0.577

The CPT reveals that male respondents with a positive family history show the highest probability of frequent interference ($P(\text{Sometimes}) = 0.577$, $P(\text{Often}) = 0.180$), while male respondents without a family history show the highest probability of no interference ($P(\text{Never}) = 0.358$). Female respondents show less variation across family history states, suggesting family history has a stronger moderating effect for male respondents.

Inference on the restricted BN further illustrates its practical utility: an individual with a family history of mental illness whose employer provides benefits and care options has a predicted treatment probability of 0.747. An individual who perceives mental health consequences at work and lacks supervisor and peer support shows $P(\text{Sometimes}) = 0.454$ and $P(\text{Often}) = 0.153$ for work interference.

Predictive performance of the restricted BN on the full dataset yielded macro $F1 = 0.471$ and $AUC = 0.771$. The $F1$ is comparable to the best baseline ML model (0.470); the AUC of 0.771 exceeds the ML baselines (maximum 0.701), but given the evaluation asymmetry both comparisons should be interpreted cautiously.

4.1.4 SHAP Analysis

SHAP values were computed for the Random Forest model using a one-vs-rest decomposition across the three classes.

The dominant predictor across all classes is `family_history`. High values produce a strong positive push toward Class 2 (interference with treatment) and a corresponding negative push away from Class 0 (no interference), consistent with the BN finding that `family_history` is a direct parent of `work_interfere`.

`care_options_Yes` and `benefits_Yes` are consistently among the top predictors. Notably, these employer-level features become the leading predictors for Class 1 (interference without treatment), where `family_history` drops to third position - suggesting that employer provisions are most relevant for distinguishing respondents experiencing interference but not yet seeking treatment.

Figure 4.3 shows the SHAP bar plot for Class 2 and Figure 4.4 the beeswarm. High `family_history` values produce large positive SHAP contributions toward Class 2 while low values push strongly away - the clearest directional effect in the dataset. `care_options_Yes` and `benefits_Yes` show a similar pattern. `Gender_female` produces a modest positive push toward Class 2, consistent with

research on gender differences in help-seeking. The anonymity features show mixed directional effects across individuals.

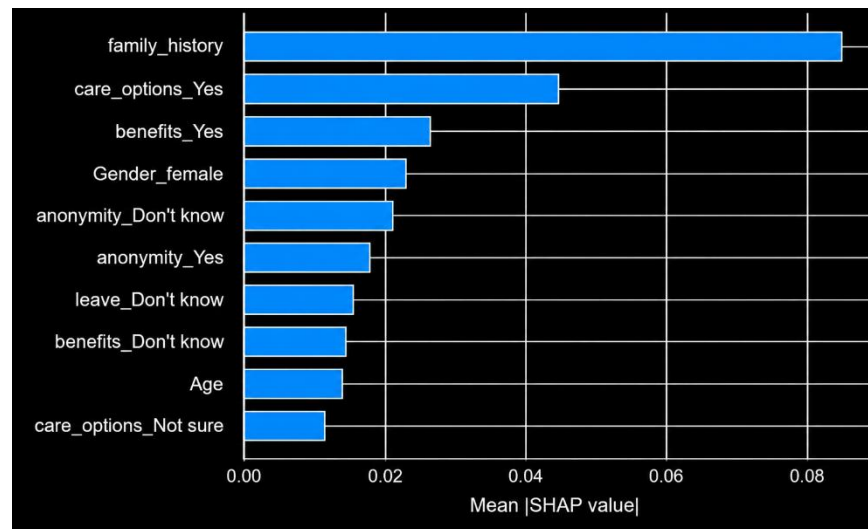


Figure 4.3 - SHAP Bar Plot: Class 2 for MH Tech Dataset

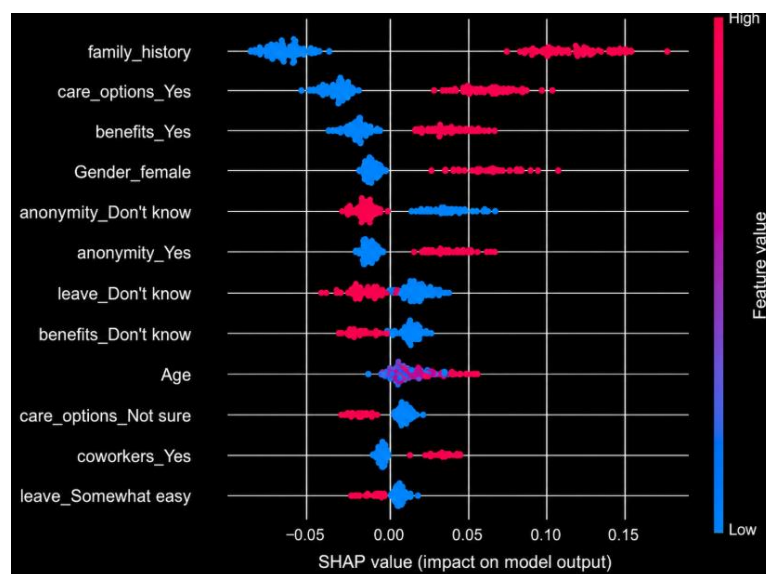


Figure 4.4 - SHAP Beeswarm: Class 2 (interference + treatment) for MH Tech Dataset

The overall SHAP profile is characterised by the dominance of a single background variable (family_history) over employer-level features, with implications for workplace mental health interventions discussed in Chapter 6.

4.2. Student Depression

4.2.1 Baseline ML Results

The four baseline models were evaluated on the binary *Depression_yes* target using five-fold stratified cross-validation. Results are reported in Table 4.4.

Table 4.4. Student Depression - Baseline ML cross-validated performance.

Model	CV Macro F1	CV AUC
Logistic Regression	0.841	0.921
Decision Tree	0.822	0.901
Random Forest	0.837	0.917
XGBoost	0.841	0.921

Performance across all baseline models is substantially higher than for MH Tech, with macro F1 values ranging from 0.822 to 0.841. Logistic regression and XGBoost are tied for the highest F1 (0.841) and AUC (0.921). Random Forest performs only marginally below these, while the decision tree again underperforms relative to the ensemble methods.

The strong performance of logistic regression, matching the best ensemble model, indicates that the predictive structure in this dataset is well captured by linear decision boundaries. This is consistent with the concentration of signal in a small number of dominant features - in particular, suicidal ideation - whose effect on the target is large and approximately monotone.

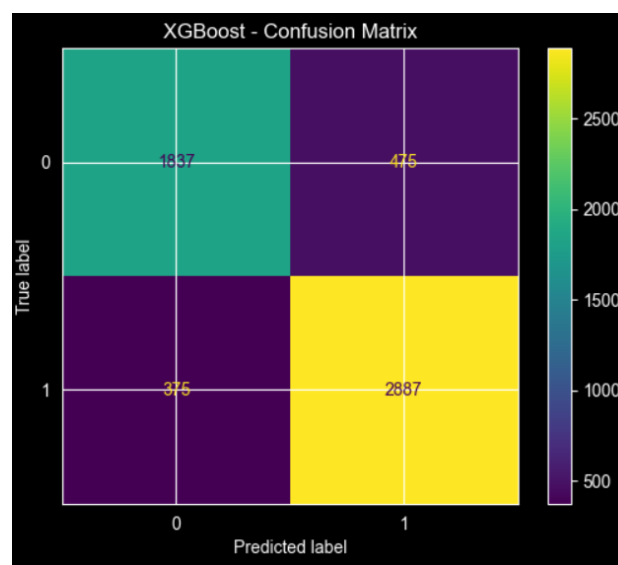


Figure 4.5 - Confusion Matrix: Best baseline model (XGBoost) for Student Depression Dataset

In Figure 4.5, the confusion matrix for XGBoost reveals that the model correctly classified 1,837 non-depressed students and 2,887 depressed students, with 475 false positives and 375 false negatives. The model demonstrates stronger recall for the positive class (depressed), correctly identifying 88.5% of depressed students, compared to 79.5% recall for the non-depressed class. The asymmetry - more false positives than false negatives - indicates the model is biased toward predicting depression, which may reflect the mild class imbalance in the dataset (2,312 non-depressed versus 3,262 depressed) as well as the strong influence of suicidal ideation as a feature, pulling borderline cases toward the positive class. In a clinical screening context this trade-off is arguably acceptable, as failing to identify a depressed student carries greater cost than a false alarm, though this is discussed further in Chapter 6.

4.2.2 Hybrid ML Results

Hybrid results are reported in Table 4.5.

Table 4.5. Student Depression - Hybrid ML cross-validated performance.

Model	CV Macro F1	CV AUC
RF Selection → LogReg	0.839	0.921
XGB Selection → LogReg	0.840	0.921
Stacking (LogReg meta)	0.840	0.921

No hybrid configuration improves on the best baseline. The maximum hybrid F1 is 0.840, one thousandth below the XGBoost baseline of 0.841, and AUC is identical across all configurations at 0.921. The cross-dataset pattern of hybrid model gains is discussed in Chapter 5.

4.2.3 BN Results

The unconstrained BN learned a graph containing 14 edges, of which 11 were evaluated as implausible - an implausibility rate of 79%, the highest observed across all five datasets. The high implausibility rate reflects the algorithm encoding strong associations between suicidal ideation, depression, and co-occurring variables in theoretically reversed directions.

The restricted BN, estimated after applying the forbidden edges list, yielded a BIC cost of -0.29% , indicating that domain constraints impose a negligible fitness cost while substantially improving structural plausibility.

The restricted BN identified the following direct parents of Depression:

- Parents of Depression: Academic Pressure, Financial Stress

This is a parsimonious and theoretically coherent structure. Academic pressure and financial stress are well-established antecedent risk factors for depression in student populations, and their identification as direct parents - rather than suicidal ideation, which is a co-symptom - reflects the value of domain constraints in recovering substantively meaningful structure from proxy-dominated data.

Table 4.6 shows selected rows from the full CPT for Depression under the restricted BN. Academic Pressure is coded 0–5 and Financial Stress 1–5.

Table 4.6. Selected CPT rows for Depression (restricted BN) - $P(\text{Depression} = \text{yes} \mid \text{Academic Pressure}, \text{Financial Stress})$.

Academic Pressure	Financial Stress	$P(\text{Depression} = \text{yes})$
0	1	0.125
0	5	0.650
3	3	0.591
5	1	0.679
5	5	0.954

The CPT reveals a clear monotone pattern: depression probability increases with both academic pressure and financial stress, reaching 0.954 when both are at their maximum. This is confirmed by inference on the restricted BN: a student with maximum academic pressure, maximum financial stress, and low study satisfaction has a predicted depression probability of 0.954 (Query 5). By contrast, a student with a history of suicidal ideation and family history of mental illness - both co-symptom variables excluded from the direct parent set - yields $P(\text{Depression} = \text{yes}) = 0.790$ (Query 4), illustrating that the restricted BN recovers antecedent risk structure rather than symptom co-occurrence.

Predictive performance of the restricted BN on the full dataset was macro F1 = 0.826, AUC = 0.906. The gap between the BN and the best ML model (0.841 F1) is 0.015 F1 points, discussed further in Chapter 5.

4.2.4 SHAP Analysis

SHAP values were computed for XGBoost, which matched logistic regression for the highest baseline F1 while providing richer non-linear structure for SHAP decomposition.

Figure 4.6 shows the mean absolute SHAP values for all features. The most important feature by a substantial margin is suicidal ideation, with a mean absolute SHAP value of approximately 1.22. This dwarfs the second and third most important features: Academic Pressure (mean $|\text{SHAP}| \approx 0.95$) and Financial Stress (mean $|\text{SHAP}| \approx 0.65$). All directional effects are consistent with theory: high suicidal ideation, high academic pressure, and high financial stress all push predictions toward depression, while low values of each push predictions away.

Beyond the top three features, the remaining predictors contribute modest but consistent signal. Work/Study Hours (≈ 0.25) and Dietary Habits_Unhealthy (≈ 0.20) are positively associated with depression risk, while Study Satisfaction and Dietary Habits_Healthy exert modest protective effects. Sleep Duration_Less than 5 hours (≈ 0.13) increases predicted risk, Family History of Mental Illness (≈ 0.12) contributes positively, and CGPA (≈ 0.10) is inversely associated with depression, with higher academic achievement reducing predicted risk.

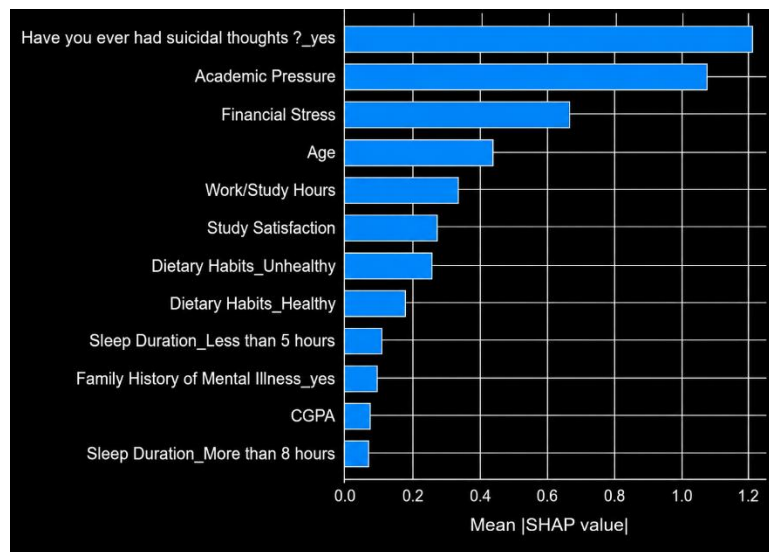


Figure 4.6 - SHAP Bar Plot: Full Model for Student Depression Dataset

Figure 4.7 shows the beeswarm plot, revealing directional patterns not visible from mean importance alone. Study satisfaction acts as a protective factor, with high values consistently producing negative SHAP contributions. Age shows a notable split: younger students are pushed toward depression while older students are pushed away. High work and study hours similarly increase predicted depression risk.

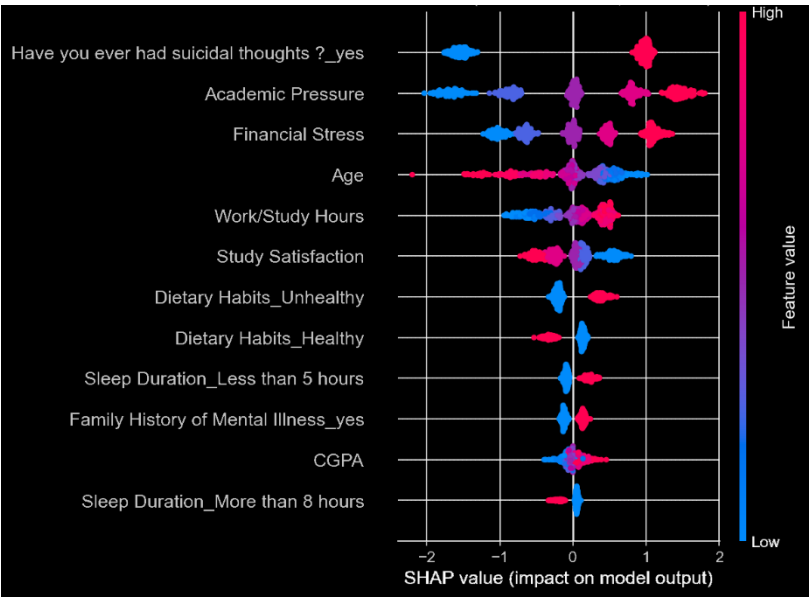


Figure 4.7 - SHAP Beeswarm: Full Model for Student Depression Dataset

4.2.5 Sensitivity Analysis

The best ML model (XGBoost) was re-run with suicidal ideation excluded from the feature set. Results are reported in Table 4.7.

Table 4.7. Student Depression - Sensitivity analysis (without suicidal ideation).

Model	CV Macro F1 (full)	CV Macro F1 (no suicidal ideation)	Δ F1
XGBoost	0.841	0.791	-0.050
CV AUC (XGBoost)	0.921	0.873	-0.048

Removing suicidal ideation reduces macro F1 by 0.050 and AUC by 0.048, yet residual performance remains substantially above chance (F1 = 0.791, AUC = 0.873), confirming genuine antecedent signal in the remaining features. As shown in Figure 4.8, Academic Pressure emerges

as the dominant predictor after exclusion, followed by Financial Stress - an ordering closely aligned with the BN's learned parent structure and discussed further in Chapter 6.

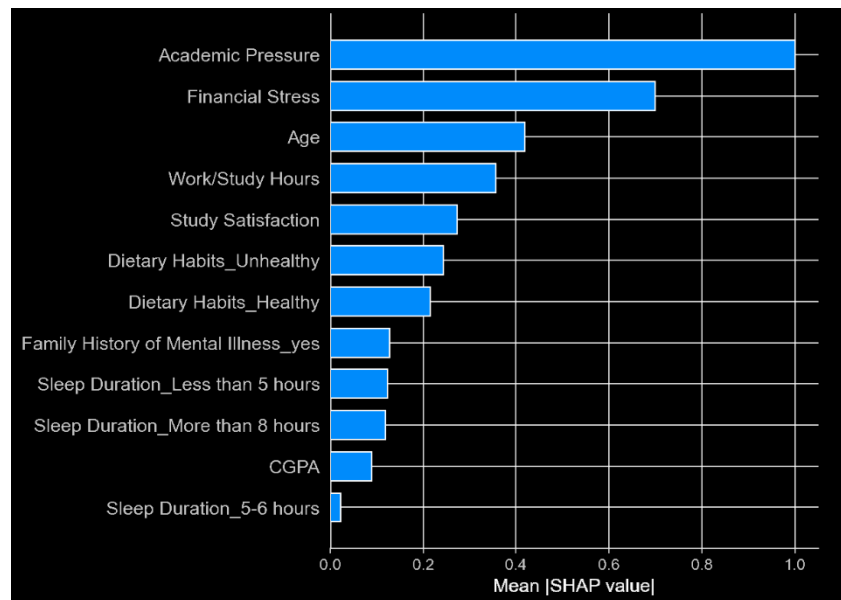


Figure 4.8 - SHAP Bar Plot: Sensitivity Analysis without Suicidal Ideation for Student Depression

4.3. Canadian Survey

4.3.1 Baseline ML Results

The four baseline models were evaluated on the continuous *Life_satisfaction* target using five-fold cross-validation. Results are reported in Table 4.8.

Table 4.8. Canadian Survey - Baseline ML cross-validated performance.

Model	CV R ²	CV RMSE
Linear Regression	0.406	1.300
Decision Tree	0.389	1.319
Random Forest	0.407	1.29
XGBoost	0.422	1.282

XGBoost achieved the best baseline performance (CV R² = 0.422, RMSE = 1.282), marginally outperforming Random Forest (R² = 0.407). Linear regression performed comparably to Random

Forest, again indicating that a substantial proportion of the predictive structure is linear. The decision tree underperformed all other baselines.

Figure 4.9 shows permutation feature importance for XGBoost. `Mental_health_state` and `Gen_health_state` dominate the importance profile by a substantial margin, accounting for the majority of predictive signal, while `Stress_level` and `Sense_belonging` contribute modestly and all remaining features contribute negligibly. The concentration of signal in self-rated health proxy variables raises questions about the nature of the predictive structure, examined in Section 4.3.5.

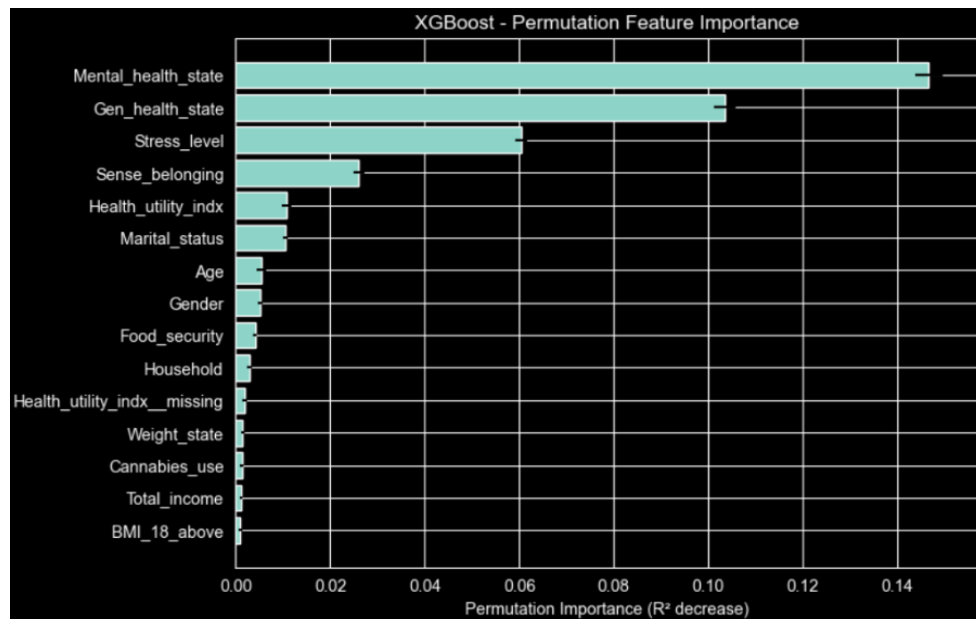


Figure 4.9 - XGBoost Permutation Feature Importance for Canadian Survey Dataset

4.3.2 Hybrid ML Results

Three hybrid configurations were evaluated: Random Forest feature selection followed by linear regression (RF → LinReg), XGBoost feature selection followed by linear regression (XGB → LinReg), and stacking with a linear regression meta-learner. Results are reported in Table 4.9.

Table 4.9. Canadian Survey - Hybrid ML cross-validated performance.

Model	CV R^2	CV RMSE
RF Selection → LinReg	0.404	1.303
XGB Selection → LinReg	0.406	1.301
Stacking (LinReg meta)	0.423	1.281

The stacking configuration marginally outperforms the best baseline ($R^2 = 0.423$ vs 0.422), a difference of $0.001 R^2$ points. The selection-then-regression pipelines produce no improvement, with both falling slightly below the linear regression baseline. The cross-dataset pattern of hybrid model gains is discussed in Chapter 5.

4.3.3 BN Results

The unconstrained BN learned 85 edges, of which 28 were evaluated as implausible - an implausibility rate of 33%, the largest raw count across all datasets, reflecting the high dimensionality and complex correlation structure of a broad national health survey.

Two restricted BN versions were estimated. The restricted BN applied the standard forbidden edges list. The strongly restricted BN additionally excluded `Mental_health_state` and `Gen_health_state` from acting as parents of the target. The BIC cost of the restricted BN was $+0.06\%$, indicating the constraints removed statistically noisy edges alongside implausible ones. The strongly restricted BN incurred a BIC cost of -4.43% , reflecting the fitness cost of excluding the proxy variables.

Both versions identified the same direct parents for `Life_satisfaction`:

- Parents of `Life_satisfaction`: `Musculoskeletal_con`, `Food_security`, `Income_source`

The stability of this structure across both constrained versions indicates robustness to proxy inclusion. All three parents are modifiable social and economic determinants of health, consistent with the OECD framework (OECD, 2020).

Inference on the restricted BN confirms the directional coherence of the learned structure. Under adverse conditions - severe food insecurity, employment income, and musculoskeletal condition present - the distribution concentrates in the mid-range (modal value 5, $P = 0.225$; $P(\text{Life_satisfaction} \leq 5) = 0.416$). Under favourable conditions - food secure, no musculoskeletal condition - the distribution shifts strongly toward higher satisfaction (modal value 8, $P = 0.342$; $P(\text{Life_satisfaction} \geq 8) = 0.577$).

Predictive performance of the restricted BN was $R^2 = 0.076$, $\text{RMSE} = 1.622$; the strongly restricted BN yielded $R^2 = 0.078$, $\text{RMSE} = 1.620$. Both are substantially below the ML baseline (best CV $R^2 = 0.422$), representing the largest predictive gap in this study. The interpretive implications are discussed in Chapter 5.

4.3.4 SHAP Analysis

SHAP values were computed for XGBoost, the best-performing baseline model.

Figure 4.10 shows the mean absolute SHAP values for all features. The two dominant predictors are Mental_health_state (mean |SHAP| ≈ 0.40) and Gen_health_state (mean |SHAP| ≈ 0.31), whose combined contribution accounts for the majority of the model's predictive variance. Beyond the proxy variables, Stress_level is the third most important predictor (≈ 0.22), with higher stress consistently pushing life satisfaction downward. Sense_belonging (≈ 0.15) appears fourth, with stronger community belonging associated with higher life satisfaction. Marital_status and Health_utility_indx (≈ 0.11) contribute modestly, all remaining features contribute negligibly.

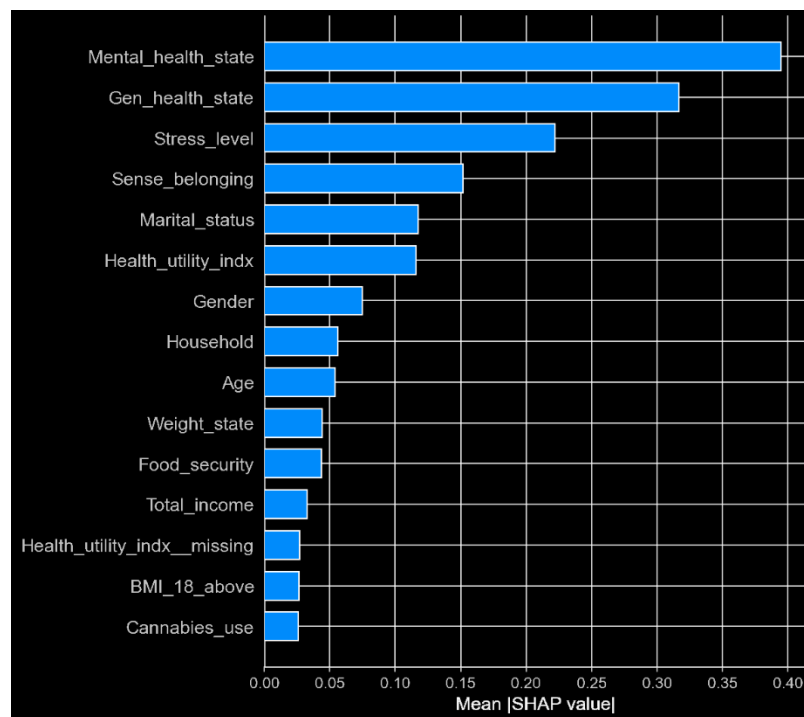


Figure 4.10 - SHAP Bar Plot: Full Model for Canadian Survey Dataset

Figure 4.11 reveals directional patterns and asymmetries not visible from mean importance alone. Poor mental and general health depress life satisfaction far more than good health elevates it - low health state values produce large negative SHAP contributions while high values cluster near zero. Food_security and Total_income show a clear positive directional effect, with higher values pushing life satisfaction upward. Gender and Household size show modest but consistent splits in the beeswarm, indicating differential effects across subgroups.

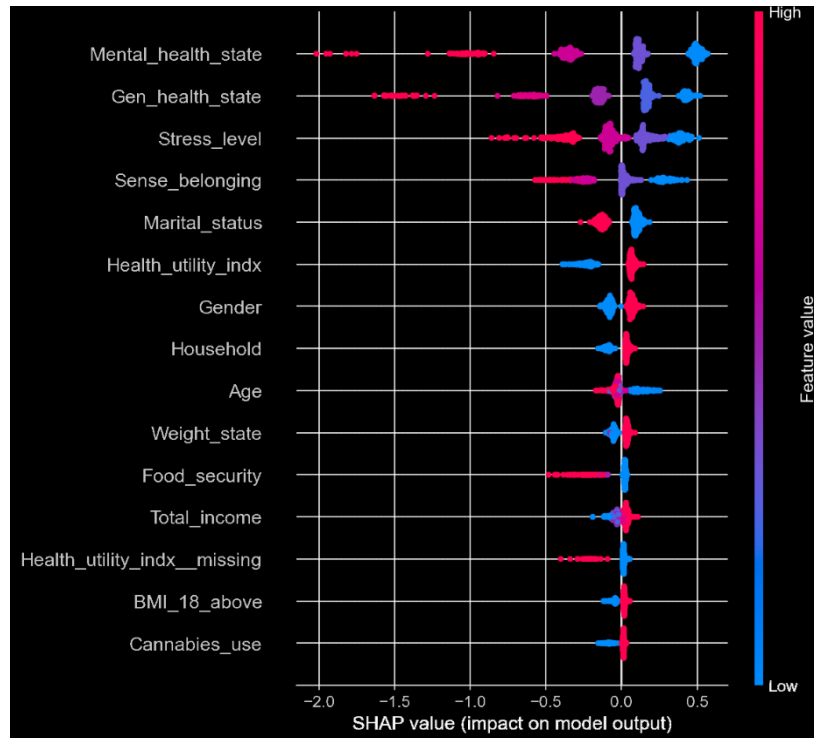


Figure 4.11 - SHAP Beeswarm: Full Model for Canadian Survey Dataset

Food_security and Income_source, identified by the BN as direct parents of the target, appear in the SHAP ranking but at lower importance than the proxy variables. This ordering reflects the statistical dominance of the proxies rather than their causal priority, a distinction the BN analysis is designed to address and discussed further in Chapter 6.

4.3.5 Sensitivity Analysis

The best ML model (XGBoost) was re-run after removing Mental_health_state and Gen_health_state from the feature set. As a result, CV R^2 dropped from 0.422 to 0.314. Removing the proxy variables produces the largest sensitivity-induced drop observed in this study ($\Delta R^2 = -0.108$), confirming that the two self-rated health variables account for approximately a quarter of total explained variance. As shown in Figure 4.12, Stress_level emerges as the dominant predictor after exclusion, with Food_security rising to a top-three position - converging with the BN's identification of food security as a direct parent of life satisfaction and discussed further in Chapter 6.

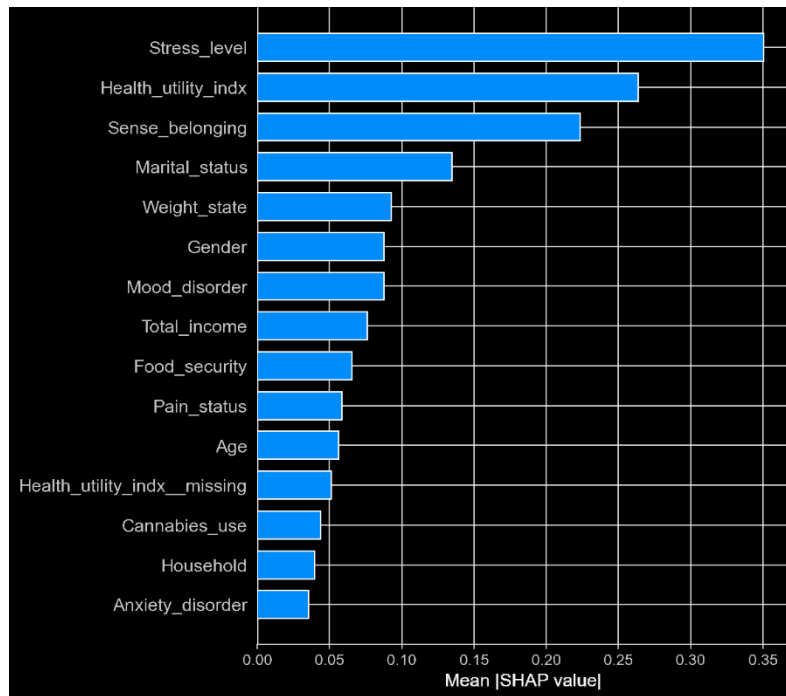


Figure 4.12 - SHAP Bar Plot: Sensitivity Analysis without Proxy Predictors for Canadian Survey Dataset

4.1. World Happiness Report

4.4.1 Synthetic Dataset - Null Result and Methodological Discussion

The WHR synthetic dataset presented an anomalous result at the first stage of modelling: all model families returned R^2 values near zero, and no hybrid or BN configuration recovered any meaningful predictive structure. As shown in Figure 4.13, the maximum absolute correlation between any feature and the target Life_Satisfaction was 0.049, consistent with a dataset in which the target variable is effectively independent of all available predictors.

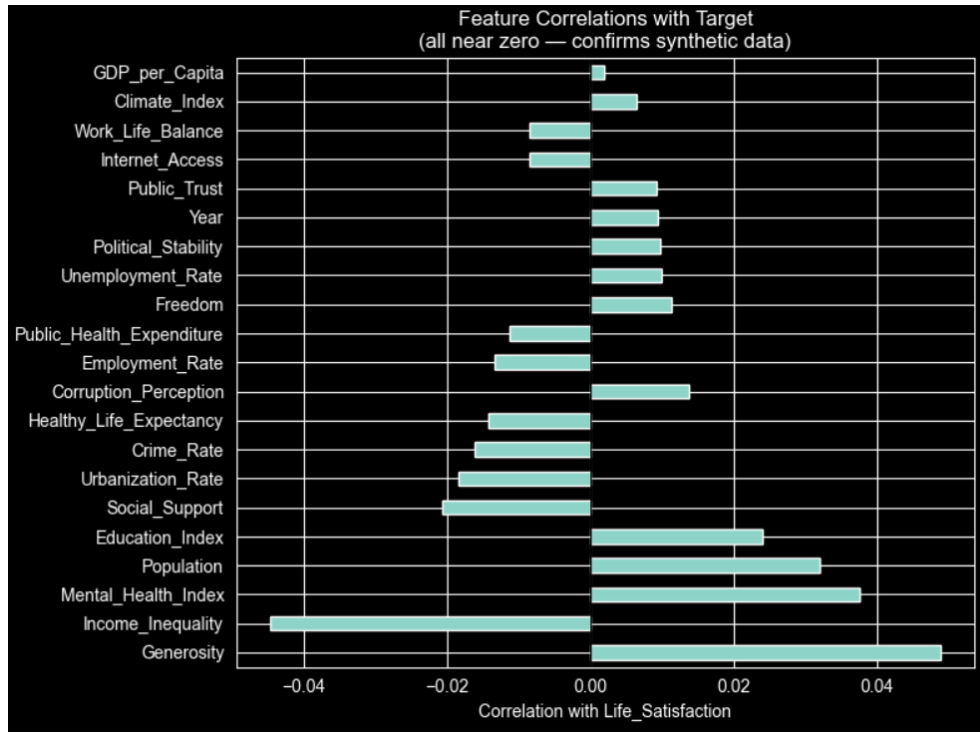


Figure 4.13 - Feature Correlation Chart for Synthetic World Happiness Report Dataset

The Bayesian Network analysis confirmed this diagnosis: both the unconstrained and restricted BN structures learned zero edges, indicating that the scoring function found no feature-target association strong enough to justify even a single directed connection.

The cross-validated evaluation framework correctly returns near-zero performance estimates, and the BN's failure to learn any structure provides complementary structural confirmation - together ruling out artefactual inflation of results. The methodological implication is that the WHR synthetic dataset cannot serve as a basis for claims about societal wellbeing predictors, and its Kaggle presentation as a real-world dataset constitutes a data quality failure. No further analysis of this dataset is reported.

4.4.2 Gallup Dataset - Results

4.4.2.1 Baseline ML Results

The four baseline models were evaluated on the continuous Life_Ladder target using five-fold cross-validation. Results are reported in Table 4.10.

Table 4.10. WHR Gallup - Baseline ML cross-validated performance.

Model	CV R ²	CV RMSE
Linear Regression	0.763	0.543
Decision Tree	0.785	0.516
Random Forest	0.843	0.442
XGBoost	0.850	0.432

The WHR Gallup dataset yields the strongest predictive performance of any dataset in this study, with XGBoost achieving $CV R^2 = 0.850$ and $RMSE = 0.432$. Random Forest is close behind ($R^2 = 0.843$). The decision tree outperforms linear regression on this dataset - a reversal of the pattern seen in the micro-level datasets - suggesting that the relationship between societal indicators and life satisfaction contains meaningful non-linearities that a single tree is better positioned to capture than a global linear model. As shown in Figure 4.14, predicted values track actual Life_Ladder scores closely across the full range, with spread widening modestly at mid-range values.

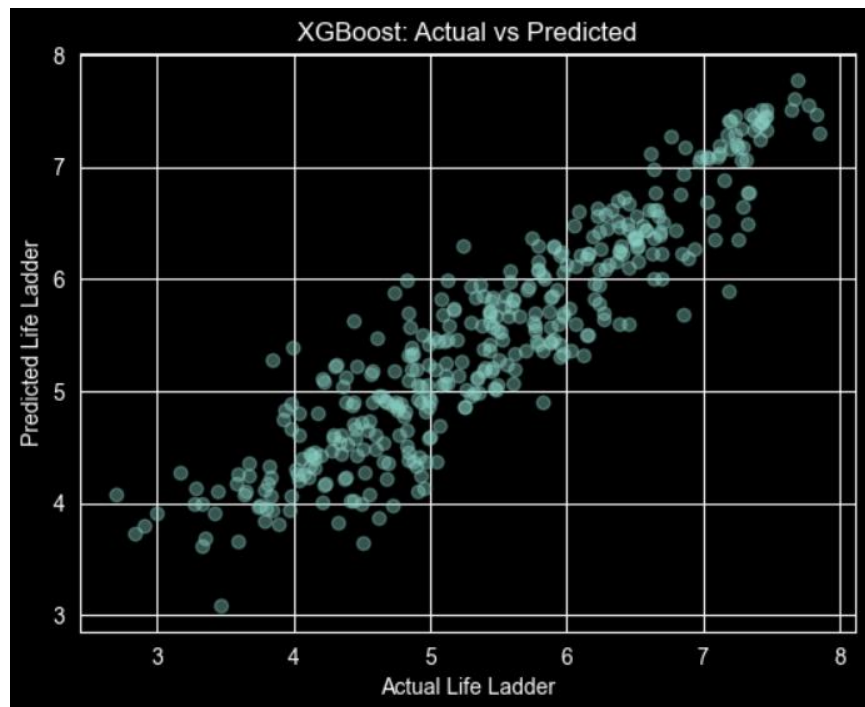


Figure 4.14 - Scatter Plot: XGBoost actual vs predicted For Gallup World Happiness Report Dataset

Figure 4.15 shows permutation feature importance for XGBoost. Log_GDP_per_capita dominates by a substantial margin (≈ 0.36 R^2 decrease), followed distantly by Healthy_life_expectancy_at_birth (≈ 0.16) and Social_support (≈ 0.09). All remaining features contribute negligibly. The macro-level comparison with Canadian Survey performance is discussed in Chapter 5.

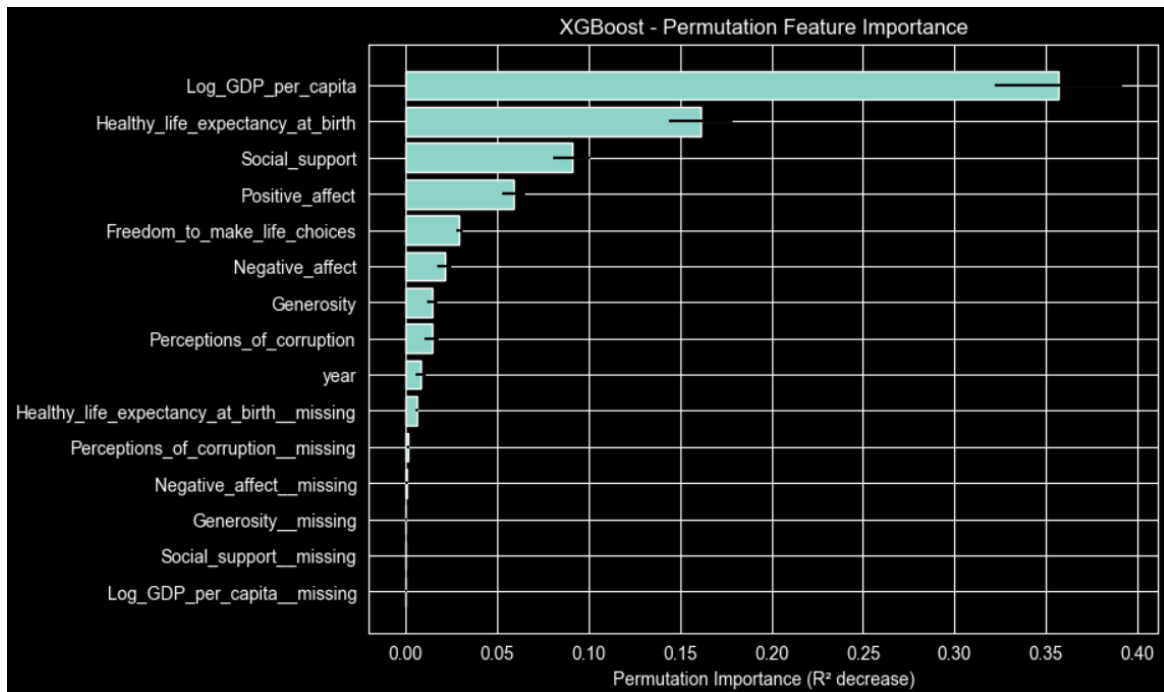


Figure 4.15 - XGBoost Permutation Feature Importance for Gallup World Happiness Report Dataset

4.4.2.2 Hybrid ML Results

Three hybrid configurations were evaluated: Random Forest feature selection followed by linear regression (RF $\rightarrow$ LinReg), XGBoost feature selection followed by linear regression (XGB $\rightarrow$ LinReg), and stacking with a linear regression meta-learner. Results are reported in Table 4.11.

Table 4.11. WHR Gallup - Hybrid ML cross-validated performance.

Model	CV R^2	CV RMSE
RF Selection $\rightarrow$ LinReg	0.759	0.547
XGB Selection $\rightarrow$ LinReg	0.755	0.551
Stacking (LinReg meta)	0.840	0.446

The hybrid models underperform the best baseline across all configurations. The selection-then-regression pipelines drop to $R^2 \approx 0.757$, a decline of approximately 0.09 R^2 points relative to the XGBoost baseline, indicating that forcing the non-linear ensemble's selected features into a linear model loses substantial predictive information on this dataset. Stacking recovers most of the gap ($R^2 = 0.840$) but still falls short of the XGBoost baseline (0.850). The cross-dataset pattern of hybrid model performance is discussed in Chapter 5.

4.4.2.3 BN Results

The unconstrained BN learned 9 edges, of which 1 was evaluated as implausible - an implausibility rate of 11%, the lowest across all datasets. The restricted BN, estimated after applying the forbidden edges list to correct the single implausible edge, achieved a BIC cost of +8.77% - the best BIC improvement observed across all datasets, indicating that the implausible edge was statistically harmful as well as theoretically implausible and its removal improved fit.

The restricted BN learned the following structure of direct relevance to the target:

- Parent of Life_Ladder: Log_GDP_per_capita (sole direct parent)
- Broader structure: Generosity $\rightarrow$ Health $\rightarrow$ GDP $\rightarrow$ Life_Ladder

This is a clean, interpretable causal chain consistent with the economic and sociological literature on national wellbeing (Helliwell et al., 2023). Table 4.12 shows the full CPT for Life_Ladder conditioned on Log_GDP_per_capita.

Table 4.12. CPT for Life_Ladder (restricted BN) - $P(\text{Life_Ladder} \mid \text{Log_GDP_per_capita})$.

Log GDP per capita	P(Very low)	P(Low)	P(Medium)	P(High)	P(Very high)
Very low	0.650	0.058	0.143	0.006	0.001
Low	0.194	0.284	0.058	0.082	0.006
Medium	0.143	0.264	0.321	0.306	0.032
High	0.012	0.083	0.280	0.421	0.180
Very high	0.001	0.006	0.032	0.179	0.744

The CPT reveals a clear monotone pattern: very low GDP is strongly associated with very low life satisfaction ($P = 0.650$), while very high GDP is strongly associated with very high life satisfaction

($P = 0.744$). The diagonal dominance confirms that `Log_GDP_per_capita` is a strong and directionally coherent direct parent of `Life_Ladder`.

Predictive performance of the restricted BN on the full dataset was $R^2 = 0.613$, $RMSE = 0.880$. Despite using a single direct parent, the BN accounts for more than 60% of the variance in national life satisfaction. The cross-dataset comparison of BN predictive performance is discussed in Chapter 5.

4.4.2.4 SHAP Analysis

SHAP values were computed for XGBoost, the best-performing model on this dataset.

Figure 4.16 shows the mean absolute SHAP values for all features. The dominant predictor is `Log_GDP_per_capita` (mean $|\text{SHAP}| \approx 0.39$), consistent with the BN's identification of GDP as the sole direct parent of `Life_Ladder` - providing convergent evidence from two independent analytical methods. `Healthy_life_expectancy_at_birth` (≈ 0.28) and `Social_support` (≈ 0.17) are the second and third most important predictors, followed by `Positive_affect` (≈ 0.16). `Freedom_to_make_life_choices`, `Negative_affect`, `Generosity`, and `Perceptions_of_corruption` contribute modestly, while all missing-indicator features contribute negligibly.

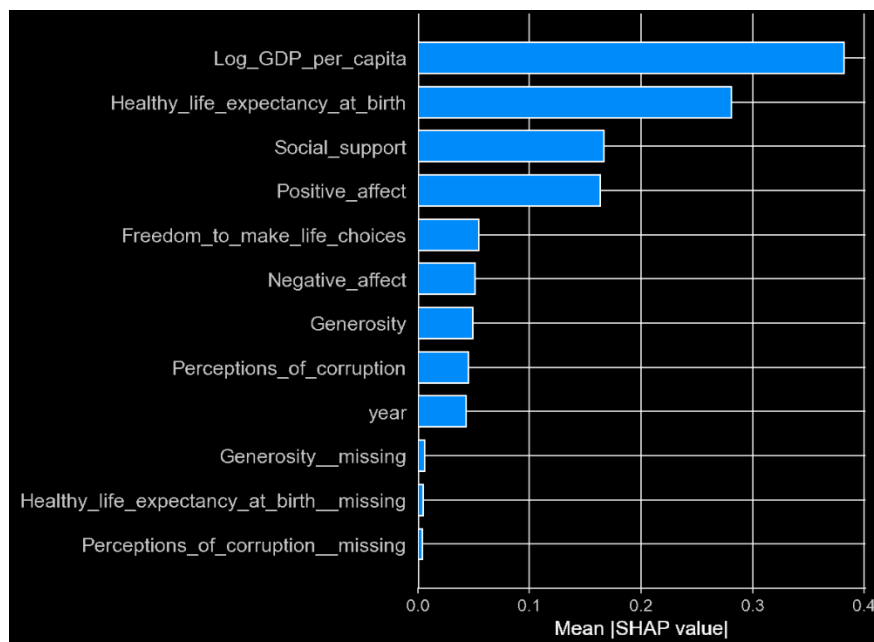


Figure 4.16 - SHAP Bar Plot: Full Model for Gallup World Happiness Report Dataset

Figure 4.17 reveals that all directional effects are consistent with the World Happiness Report theoretical framework: higher GDP, social support, life expectancy, and freedom push predictions toward higher life satisfaction, while higher negative affect and perceived corruption push predictions downward. The beeswarm shows these effects are monotone and symmetric across the feature value range, indicating the model has learned a coherent representation of the macro-level wellbeing literature. Notably, Generosity shows a wide spread in the beeswarm despite its modest mean importance, suggesting high variance in its effect across country-year observations.

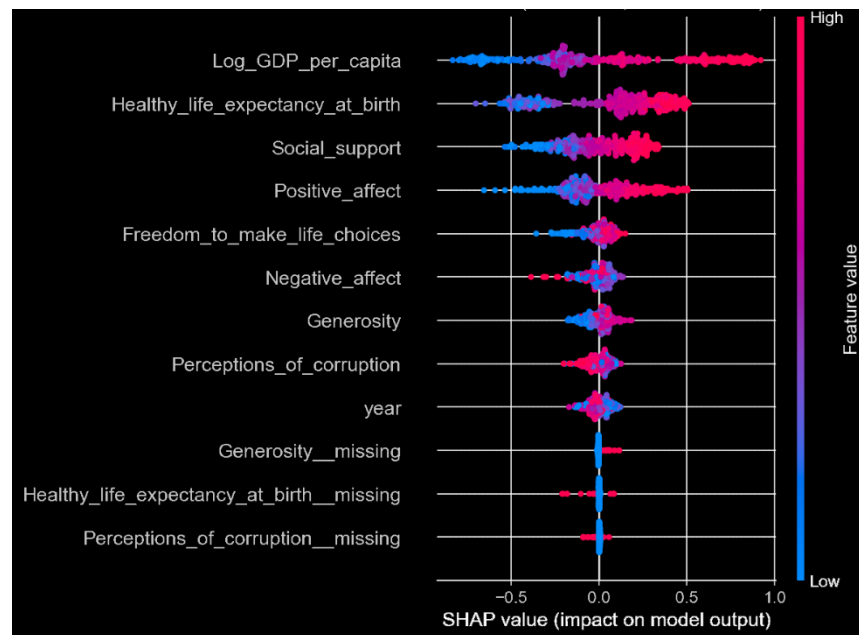


Figure 4.17 - SHAP Beeswarm: Full Model for Gallup World Happiness Report Dataset

4.4.2.5 Sensitivity Analysis

The best ML model (XGBoost) was re-run after removing positive and negative affect from the feature set. CV R^2 dropped from 0.850 to 0.832. Removing affect variables produces the smallest sensitivity-induced reduction in this study ($\Delta R^2 = -0.018$), indicating they are not functioning as dominant proxies for the Cantril scale. As shown in Figure 4.18, Log_GDP_per_capita remains the dominant predictor after exclusion, with the ordering of social support, life expectancy, and freedom preserved - discussed further in Chapter 6.

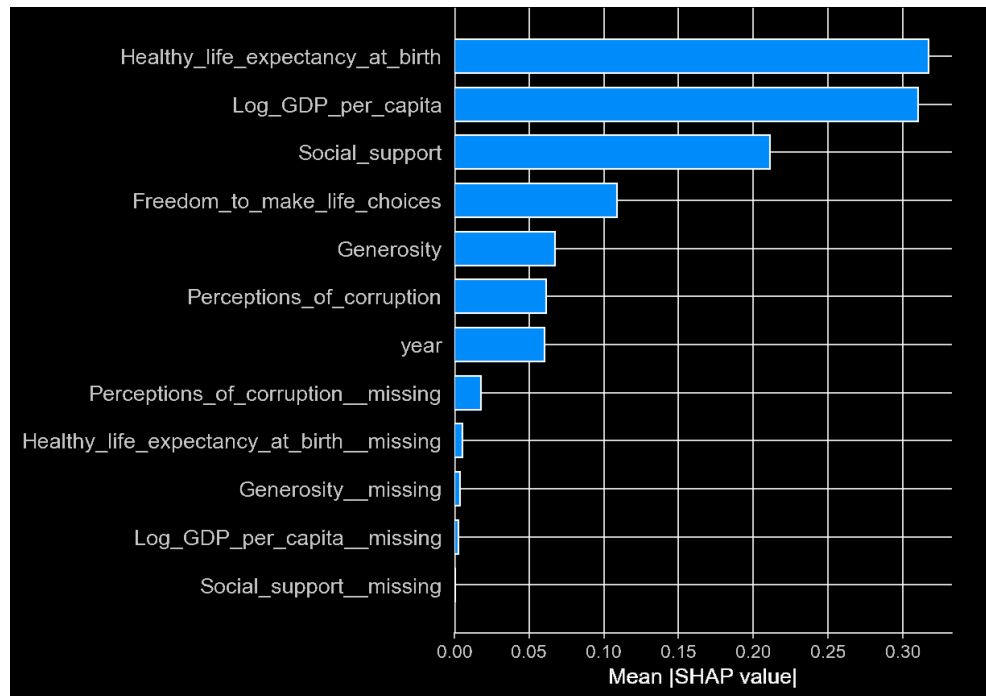


Figure 4.18 - SHAP Bar Plot: Sensitivity Analysis for Gallup World Happiness Report Dataset

Chapter 5 - Cross Dataset Analysis

5.1. Model comparison tables and figures

This section aggregates the cross-validated performance results from Chapter 4 into unified comparison tables, enabling direct assessment of how model families perform relative to one another across datasets and task types. Because the five datasets span two task types - classification and regression - results are presented in two separate tables.

5.1.1 Classification Datasets

Table 5.1 reports macro F1 and AUC for all model configurations across the two classification datasets. For Bayesian Networks, full-data metrics are reported and marked accordingly.

Table 5.1. Classification datasets - full model comparison (CV unless marked †).

Model	MH Tech F1	MH Tech AUC	Student Depression F1	Student Depression AUC
Logistic Regression	0.457	0.685	0.841	0.921
Decision Tree	0.408	0.640	0.822	0.901
Random Forest	0.470	0.696	0.841	0.921
XGBoost	0.441	0.691	0.841	0.921
RF → LogReg	0.465	0.701	0.839	0.921
XGB → LogReg	0.488	0.701	0.840	0.921
Stacking	0.460	0.700	0.840	0.921
BN (restricted) †	0.471	0.771	0.826	0.906

Several patterns are immediately apparent. First, the performance gap between the two datasets is large and consistent across all model families. Student Depression achieves F1 values in the range 0.822–0.841 regardless of model complexity, while MH Tech achieves F1 values in the range 0.408–0.488. This gap does not narrow when more complex models are applied, indicating that it reflects differences in signal quality and task difficulty between datasets rather than the limits of any particular modelling approach. The Student Depression dataset benefits from a large sample

($n = 27,870$), a binary target, and a dominant predictor; MH Tech has a more balanced signal distribution, a three-class target, and a moderate sample size.

Second, the ordering of model families is inconsistent across the two datasets. On MH Tech, the hybrid XGB $\rightarrow$ LogReg pipeline achieves the highest F1 (0.488), marginally ahead of Random Forest (0.470). On Student Depression, logistic regression and XGBoost tie at the top (0.841) and no hybrid configuration improves on them. This pattern is discussed further in Chapter 6.

Third, the BN achieves F1 near the ML baseline on both datasets - 0.471 versus the best CV F1 of 0.488 on MH Tech, and 0.826 versus 0.841 on Student Depression - despite using structurally constrained, parsimonious parent sets and being evaluated on the full dataset rather than cross-validated held-out folds. The BN AUC on MH Tech (0.771) appears to exceed all ML configurations (maximum 0.701), but this advantage is attributed to the evaluation asymmetry and should not be interpreted as a direct performance comparison.

5.1.2 Regression Datasets

Table 5.2 reports R^2 and RMSE for all model configurations across the three regression datasets. WHR Synthetic is included for completeness; its near-zero values serve as a null baseline. BN metrics are full-data and marked accordingly.

Table 5.2. Regression datasets - full model comparison (CV unless marked †).

Model	Canadian R^2	Canadian RMSE	Gallup R^2	Gallup RMSE	Synthetic R^2
Linear Regression	0.406	1.300	0.763	0.543	≈ 0.000
Decision Tree	0.389	1.319	0.785	0.516	≈ 0.000
Random Forest	0.407	1.299	0.843	0.442	≈ 0.000
XGBoost	0.422	1.282	0.850	0.432	≈ 0.000
RF $\rightarrow$ LinReg	0.404	1.303	0.759	0.547	≈ 0.000
XGB $\rightarrow$ LinReg	0.406	1.301	0.755	0.551	≈ 0.000
Stacking	0.423	1.281	0.840	0.446	≈ 0.000
BN (restricted) †	0.076	1.622	0.613	0.880	-

The most salient feature of Table 5.2 is the performance gap between the Canadian Survey and WHR Gallup: XGBoost achieves $R^2 = 0.850$ on Gallup versus 0.422 on the Canadian Survey, a difference of 0.428 R^2 points that persists across every model family and is interpreted in Section 5.3. The WHR Synthetic column confirms the null-signal diagnosis, with every model family returning effectively zero performance.

The BN predictive gap, visible in Figure 5.1, varies substantially across datasets - largest on the Canadian Survey (BN $R^2 = 0.076$ versus ML best of 0.422, a gap of 0.346) and smallest on Student Depression (BN F1 = 0.826 versus ML best of 0.841, a gap of 0.015). This variation is explored further in Section 5.2.

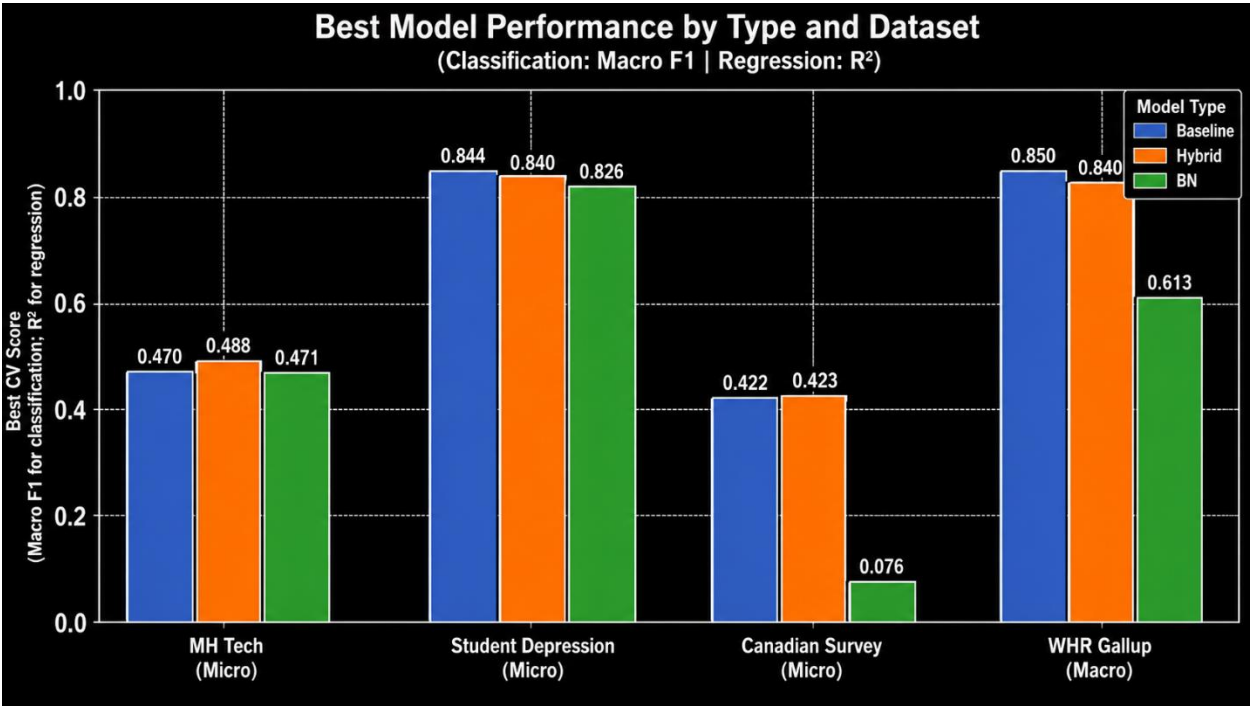


Figure 5.1 - Best Model Performance by Type and Dataset: Grouped Bar Chart

5.1.3 Hybrid Model Gains

Table 5.3 summarises the performance gain of the best hybrid configuration over the best baseline ML model on each dataset, isolating the incremental contribution of hybridisation.

Table 5.3. Hybrid model gain over best baseline - all datasets.

Dataset	Best Baseline	Best Hybrid	Metric	Gain
MH Tech	0.470 (RF)	0.488 (XGB→LR)	Macro F1	+0.018
Student Depression	0.841 (LR/XGB)	0.840 (Stacking)	Macro F1	−0.001
Canadian Survey	0.422 (XGB)	0.423 (Stacking)	CV R ²	+0.001
WHR Gallup	0.850 (XGB)	0.840 (Stacking)	CV R ²	−0.010

Hybrid gains are negligible across all four datasets, ranging from −0.010 to +0.018. The maximum gain, observed on MH Tech, is 0.018 F1 points - a result that does not constitute a practically meaningful improvement given the measurement uncertainty in cross-validated estimates. On two of the four datasets, the best hybrid underperforms the best baseline. The practical implications of this finding are discussed in Chapter 6.

5.2. BN consolidation

This section brings together the Bayesian Network results from Chapter 4 to assess what the BNs collectively reveal about the structure of wellbeing data, how the learned structures compare across datasets, and how the tension between statistical fit and domain plausibility varies with data characteristics.

5.2.1 Learned Causal Structures

Table 5.4 summarises the direct parents of the target variable identified by the restricted BN on each dataset, alongside the broader structural features of the learned graph.

Table 5.4. BN structural summary - all datasets.

Dataset	Direct parents of target	Total edges	Implausible edges (unconstrained)	Implausibility rate	BIC cost of constraints
MH Tech	Gender, family history	42	10	24%	−3.98%
Student Depression	Academic Pressure, Financial Stress	14	11	79%	−0.29%
Canadian Survey	Musculoskeletal_con, Food_security, Income source	85	28	33%	+0.06%
WHR Gallup	Log_GDP_per_capita	8	1	11%	+8.77%

The implausibility rate is inversely related to data quality and signal strength. WHR Gallup, with the strongest genuine signal, produces only 11% implausible edges in the unconstrained BN, while Student Depression, dominated by a co-symptom proxy, produces 79%. MH Tech and the Canadian Survey fall between these extremes.

The BIC cost of constraints is similarly related to data quality. For WHR Gallup, applying constraints improves BIC by +8.77%, meaning the forbidden edges list removes statistically harmful structure as well as implausible structure. For the Canadian Survey's restricted BN the cost is near zero (+0.06%), indicating constraints are orthogonal to statistical fit. For MH Tech (−3.98%) and the strongly restricted Canadian BN (−4.43%) there is a modest fitness cost.

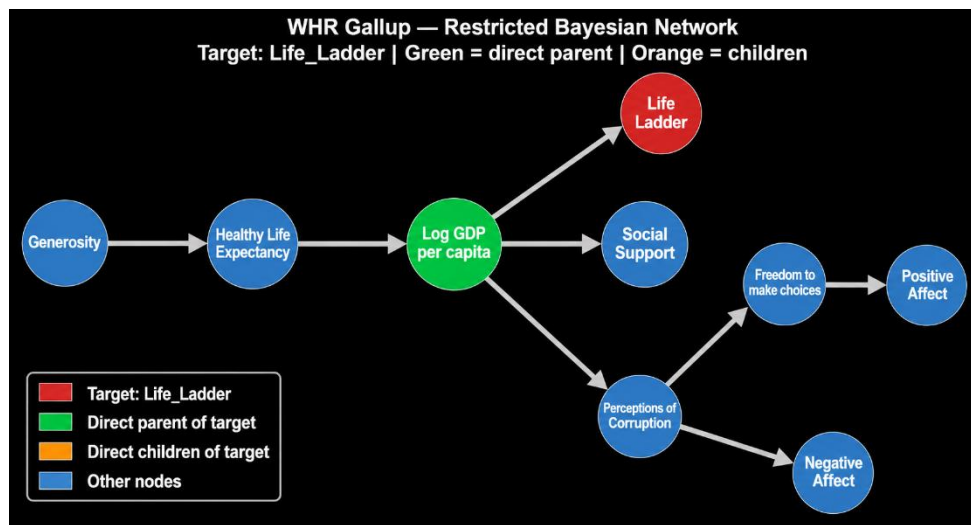


Figure 5.2 - WHR Gallup restricted BN graph (full 9-node structure)

The direct parent sets identified by the restricted BNs are parsimonious and theoretically coherent across all four datasets, converging with established theoretical frameworks: familial risk and gender as determinants of workplace mental health interference; academic and financial stress as antecedent risk factors for student depression; material conditions as determinants of individual life satisfaction; and economic development as the primary determinant of national life satisfaction. Figure 5.5 illustrates the full learned structure for WHR Gallup, the cleanest recovered graph in this study.

5.2.2 BN Predictive Performance Relative to ML

Table 5.5 summarises the predictive gap between the restricted BN and the best ML baseline on each dataset.

Table 5.5. BN vs best ML baseline - predictive gap and structural parsimony.

Dataset	Best ML (CV)	BN (full data)	Gap	BN direct parents
MH Tech	F1 = 0.488	F1 = 0.471	-0.017	2
Student Depression	F1 = 0.841	F1 = 0.826	-0.015	2
Canadian Survey	R ² = 0.422	R ² = 0.076	-0.346	3
WHR Gallup	R ² = 0.850	R ² = 0.613	-0.237	1

The predictive gap is smallest on the two classification datasets, where the BN achieves near-parity despite using only two direct parents. On the regression tasks the gap is substantial, particularly for the Canadian Survey, reflecting both the larger feature set available to ML models and the cost of discretising a continuous target for BN inference. The appropriate evaluative frame for BN regression performance is not how closely it approaches the ML ceiling, but whether the structure it encodes is causally meaningful - a trade-off examined in Chapter 6.

5.3. Micro vs macro comparison

A primary axis of comparison is the distinction between micro-level datasets - individual survey responses - and the macro-level WHR Gallup dataset, which aggregates responses to the country-year level.

5.3.1 Predictive Performance

Table 5.6 compares the best cross-validated R^2 achieved on the two regression datasets.

Table 5.6. Micro vs macro regression performance - best ML model.

Dataset	Level	Best CV R^2	Best CV RMSE	Target scale
Canadian Survey	Micro	0.422	1.282	0–10 (individual)
WHR Gallup	Macro	0.850	0.432	0–10 (country mean)

The macro dataset achieves more than twice the R^2 of the micro dataset using the same model family and overlapping predictor concepts. The performance difference is not attributable to target scale or predictor domain, but to the structural difference between individual and aggregate data: at the individual level, measurement error, transient mood states, and idiosyncratic circumstances impose an irreducible ceiling on achievable R^2 . At the country level these fluctuations average out, leaving a cleaner signal in which GDP, social institutions, and health infrastructure explain systematic cross-national variation.

5.3.2 Model Family Ordering

On the Canadian Survey the decision tree underperforms linear regression ($R^2 = 0.389$ vs 0.406) and the gap between linear regression and XGBoost is narrow (0.406 vs 0.422). On WHR Gallup the decision tree outperforms linear regression (0.785 vs 0.763) and the gap between linear regression and XGBoost is large (0.763 vs 0.850). This pattern suggests that non-linear relationships and feature interactions are more prevalent and more learnable in macro-level data, where structured non-linearities such as diminishing returns to GDP benefit tree-based models. The practical implications are discussed in Chapter 6.

5.3.3 BN Structural Alignment with Theory

Both micro and macro BNs identify theoretically coherent parent sets, but alignment is tighter at the macro level. The WHR Gallup BN structure - Generosity $\rightarrow$ Health $\rightarrow$ GDP $\rightarrow$ Life_Ladder - maps cleanly onto the World Happiness Report theoretical framework (Helliwell et al., 2023). The micro-level BNs produce coherent but more modestly supported structures, with high implausibility rates in their unconstrained versions indicating that individual survey data is

substantially more ambiguous with respect to causal direction. Figure 5.3 illustrates the contrast between individual-level and macro-level causal pathways across the study's datasets.

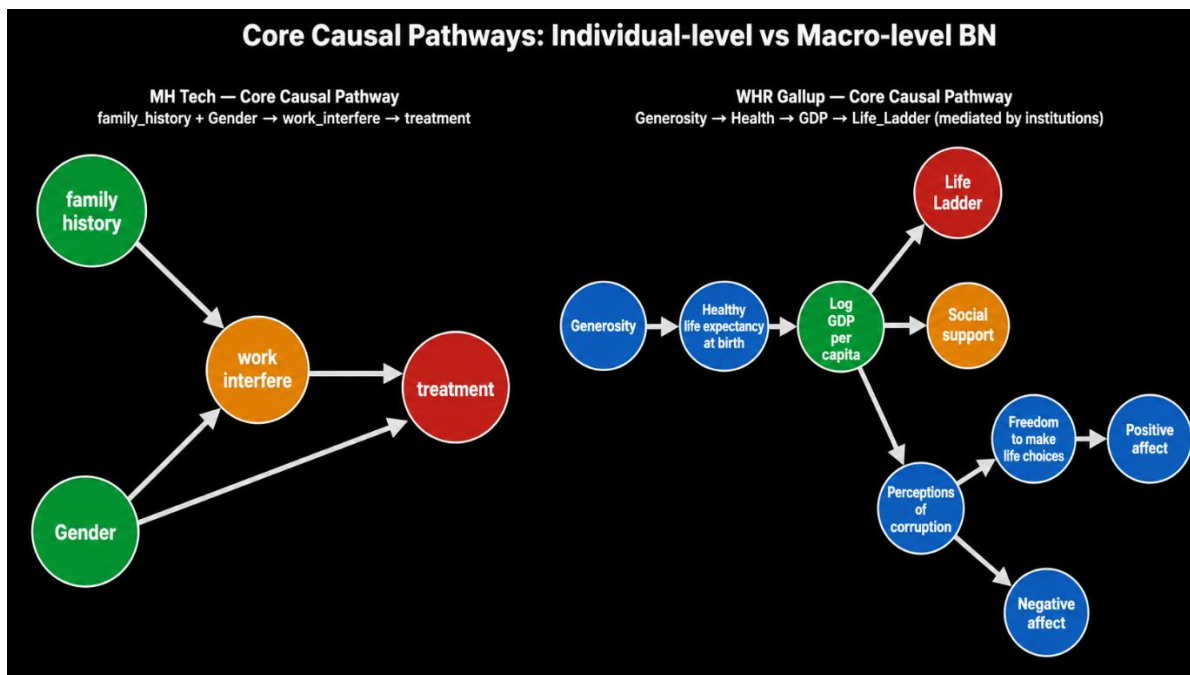


Figure 5.3 - Core causal pathways: individual vs macro level (side-by-side)

5.4. SHAP cross-dataset patterns

Across the four substantive datasets, SHAP analysis reveals a set of recurring patterns that cut across measurement level, task type, and predictor domain.

5.4.1 Proxy Dominance

In two of the four datasets, a single feature dominates the SHAP importance ranking by a margin large enough to substantially reshape the model's behaviour. In the Student Depression dataset, suicidal ideation accounts for the majority of predictive variance, dwarfing all other features. In the Canadian Survey, the combined contribution of `Mental_health_state` and `Gen_health_state` reduces all other predictors to secondary status.

In both cases, the dominant feature is not a genuine upstream cause of the outcome but an evaluatively similar proxy. The sensitivity analyses confirm this interpretation: removing the proxies reduces performance by 0.050 F1 points (Student Depression) and 0.108 R^2 points (Canadian Survey) but does not collapse prediction, and the residual SHAP rankings converge on

theoretically meaningful antecedent predictors - academic and financial stress in the former, and stress level and food security in the latter. The implications for interpreting SHAP rankings are discussed in Chapter 6.

5.4.2 Convergence Between SHAP and BN Parents

A consistent pattern across datasets is that the variables identified by the restricted BN as direct parents of the target are also among the top predictors in the post-proxy-exclusion SHAP rankings, even though the two methods are methodologically independent. This convergence is observed in three of the four datasets:

In the Student Depression dataset, the BN identifies Academic Pressure and Financial Stress as direct parents; after removing suicidal ideation, these become the top two SHAP predictors. In the Canadian Survey, the BN identifies Food_security and Income_source as direct parents; after removing the proxy health variables, Food_security emerges as a top-three SHAP predictor. In WHR Gallup, the BN identifies Log_GDP_per_capita as the sole direct parent; it is also the dominant SHAP predictor both with and without affect variables included.

This convergence suggests that SHAP applied to a well-specified model from which proxy variables have been excluded and the BN applied with domain constraints tend to agree on what matters - strengthening the evidential basis for the identified predictors.

5.4.3 Directional Consistency and Asymmetric Effects

Across all four datasets, the directional effects identified by SHAP are consistent with theoretical expectations without exception. A recurring asymmetry is also present: negative predictor values (poor health, high stress, low income) produce larger SHAP contributions than equivalent positive values, consistent with negativity bias in subjective wellbeing research (Diener, 1984), and the effect of GDP shows diminishing returns at the upper end of the distribution. These asymmetries are invisible to linear models and would be missed by coefficient-based interpretation alone.

Chapter 6 - Discussion

6.1. Answers to each RQ

This section addresses each of the five research questions set out in Chapter 1 directly and systematically, drawing on the results reported in Chapters 4 and 5.

6.1.1 RQ1: How Do Different Modelling Approaches Perform When Predicting Wellbeing at Individual Versus Societal Levels?

The results provide a clear and consistent answer: predictive performance is substantially higher at the societal level than at the individual level, and this difference is robust across all model families tested. The best individual-level regression model (XGBoost on the Canadian Survey) achieves $CV R^2 = 0.422$, while the best societal-level model (XGBoost on WHR Gallup) achieves $CV R^2 = 0.850$. This gap of $0.428 R^2$ points persists regardless of whether simple linear models, ensemble methods, or hybrid pipelines are used, indicating that it reflects a structural property of the data rather than a modelling artefact.

At the individual level, measurement error, transient mood effects, and idiosyncratic circumstances impose an irreducible noise ceiling that no model can overcome; at the country level these fluctuations average out, leaving a cleaner signal in which GDP, social institutions, and health infrastructure account for systematic cross-national variation.

Among individual-level datasets, performance varies with signal concentration: Student Depression achieves $CV F1 = 0.841$ driven largely by a co-symptom proxy, while MH Tech yields a ceiling of $0.488 F1$ reflecting a more typical distributed-signal prediction problem. At the macro level, tree-based ensembles substantially outperform linear regression (R^2 gap of 0.087), while at the individual level the gap between linear regression and XGBoost is narrow ($0.016 R^2$), suggesting that predictive structure in individual survey data is predominantly linear.

6.1.2 RQ2: How Does Interpretability Differ Between Regression, ML, and Bayesian Networks?

The three model families provide fundamentally different forms of interpretability, and these differences are consequential for how findings should be communicated and used.

Linear and logistic regression provide coefficient-based interpretability: every predictor has a single estimated effect size with a direction and magnitude directly readable from the model output. This interpretability is unconditional - it does not require post-hoc analysis tools and is available for any prediction. However, coefficients assume linear, additive effects and provide no information about non-linearities, interactions, or heterogeneity across subgroups.

Machine learning models, particularly XGBoost, achieve higher predictive accuracy but sacrifice direct coefficient interpretability. SHAP analysis partially recovers this by decomposing each prediction into additive feature contributions, providing both global importance rankings and local, instance-level explanations. However, SHAP interpretability is post-hoc and attribution-based rather than structural: it explains what the model has learned to associate with the outcome, not why those associations exist or what causal pathway they reflect.

Bayesian Networks provide a third form of interpretability that is qualitatively different from both coefficient tables and SHAP rankings: a directed graph encoding conditional independence relationships among all variables. The BN does not merely rank predictors by importance; it makes explicit claims about which variables directly influence which others, and in what direction. The restricted BN parent sets in this study - family history and gender for workplace mental health interference; academic pressure and financial stress for student depression; food security and income for life satisfaction; GDP for national life satisfaction - represent structurally explicit causal hypotheses that coefficient tables and SHAP rankings do not provide.

The trade-off is that BN structural interpretability comes at a cost to predictive coverage. Whether this is worthwhile depends on the research goal: maximum predictive accuracy favours ML with SHAP; identifying a parsimonious set of directly influential factors to guide intervention or policy favours the BN structure.

6.1.3 RQ3: Are Bayesian Networks Better at Capturing Plausible Causal Structures in Wellbeing Data?

The evidence from this study is nuanced and dataset-dependent. When applied to high-quality, moderate-dimensional data with strong genuine signal - as in WHR Gallup - the BN recovers a largely plausible structure even without domain constraints (implausibility rate of 11%), and the application of constraints improves both structural validity and BIC score simultaneously.

When applied to individual-level survey data characterised by proxy variables, high dimensionality, or noisy correlation structures, the unconstrained BN performs poorly as a causal discovery tool. The 79% implausibility rate on the Student Depression dataset and the 33% rate on the Canadian Survey indicate that the structure learning algorithm, without domain guidance, frequently encodes statistically convenient but theoretically incoherent causal directions. This is not a failure specific to BNs: any unsupervised causal discovery method applied to observational survey data faces the fundamental problem that correlation does not uniquely identify causal direction, and proxy variables produce associations indistinguishable from genuine causal pathways by statistical criteria alone.

The key finding is therefore that BNs are capable of capturing plausible causal structures, but this capacity is conditional on the application of expert knowledge constraints. The restricted BNs in this study consistently identify theoretically coherent parent sets, and the convergence of these parent sets with the post-proxy-exclusion SHAP rankings provides independent corroboration that the BN structures are not artefactual. Expert knowledge is an integral component of the BN approach: it provides a formal framework for incorporating domain constraints and then learning the remaining structure from data, which is precisely what observational causal inference requires (Pearl, 2009; Morgan and Winship, 2015).

The BIC cost of constraints serves as a practically useful diagnostic: a large BIC improvement under constraints indicates that statistical and domain criteria reinforce one another, while a large BIC cost indicates that the domain-plausible structure is genuinely less statistically supported and that interpretive caution is warranted.

6.1.4 RQ4: Can Bayesian Networks Reveal Causal Structures That ML Models Cannot?

Yes, with an important qualification about what is meant by revealing causal structure. Machine learning models, including XGBoost with SHAP, can identify which variables are statistically important and whether their effects are positive or negative, but they cannot distinguish between a variable that directly causes the outcome and a variable that is associated with the outcome through a common cause, a mediating pathway, or a proxy relationship.

The BN, by contrast, encodes a specific claim about the conditional independence structure of the joint distribution: the parents of a node are those variables that directly influence it, with all other associations explained by indirect pathways or shared causes. When the restricted BN identifies

food security and income source as direct parents of life satisfaction in the Canadian Survey while placing mental health state and general health state elsewhere in the graph, it is making a structural claim that SHAP cannot make - namely, that material conditions have a direct influence on life satisfaction that is not fully mediated by health status.

Similarly, the BN's identification of the Generosity $\rightarrow$ Health $\rightarrow$ GDP $\rightarrow$ Life_Ladder chain on WHR Gallup reveals a mediating pathway structure that neither a coefficient table nor a SHAP ranking surfaces. The SHAP analysis ranks GDP as the most important predictor and generosity among the weakest, which is accurate in terms of direct statistical contribution but obscures the indirect pathway through which generosity operates. The BN makes this pathway explicit.

The qualification is that BNs reveal the conditional independence structure of the data, not ground-truth causal structure. On observational data, a DAG is identifiable only up to a Markov equivalence class, and the Hill Climb Search algorithm makes choices within this equivalence class based on scoring rather than genuine causal identification. The structures identified in this study are therefore best interpreted as causally plausible hypotheses consistent with both the data and domain knowledge, rather than confirmed causal facts.

6.1.5 RQ5: How Robust Are These Models to Missing Data, Noise, and Correlated Predictors?

Robustness varies systematically across model families and data characteristics.

Regarding missing data, the preprocessing strategy of adding binary missingness indicators alongside imputed values proved effective across all ML pipelines. The missingness indicators on the WHR Gallup dataset carried non-trivial SHAP importance, indicating that the pattern of missingness itself is informative - countries with missing data on certain variables are not randomly distributed. The BN pipeline's strategy of preserving missing values as a distinct state is theoretically cleaner but practically constrained by the sparsity it introduces into conditional probability tables for rarely-occurring missing patterns.

Regarding noise and weak signal, the WHR Synthetic case study demonstrates that the cross-validated ML pipeline correctly returns near-zero performance on a null-signal dataset, and the BN correctly learns zero edges. This robustness to null signal is a desirable property: a pipeline that overfits to noise would produce positive performance estimates and potentially misleading

structural outputs, and the consistent near-zero results across all methods provide confidence that positive results elsewhere reflect genuine signal.

Regarding correlated predictors, multicollinearity is most acute in the Canadian Survey, where a large number of health and socioeconomic variables share substantial variance. Linear regression is most sensitive to this, as it can produce unstable coefficient estimates when predictors are highly correlated. Ensemble methods are more robust because individual trees consider random feature subsets, effectively averaging over the correlated feature space. The BN handles correlated predictors through the conditional independence framework: if two features are correlated only because they share a common cause, the BN will encode that common cause rather than creating a direct edge between them.

6.2. Prediction-explanation trade-off

The prediction-explanation trade-off - the tension between models that predict well and models that explain clearly - is a central theme in the methodology and philosophy of statistical modelling (Breiman, 2001; Shmueli, 2010). The results of this study provide a concrete, multi-dataset illustration of how this trade-off manifests in applied wellbeing research.

The hybrid model findings speak most directly to the prediction side of the trade-off. The consistent failure of hybrid configurations to outperform the best single model - with gains ranging from -0.010 to $+0.018$ across all datasets - implies that there is no stable middle ground in which interpretability is recovered without predictive cost. The selection-then-regression pipelines reduce the feature set to those identified as important by ensemble methods, then fit a transparent linear model on that subset. In theory this should combine the feature relevance knowledge of the ensemble with the interpretability of the linear model. In practice it delivers neither the full predictive power of the ensemble nor the correct coefficient estimates of a properly specified linear model.

The explanation side is illustrated most clearly by the BN results. The Canadian Survey BN, constrained to a causally plausible three-parent structure, achieves $R^2 = 0.076$ against an ML baseline of 0.422 . This is not a failure of the BN as a modelling tool; it is the cost of prioritising structural coherence over predictive coverage. A model that correctly identifies food security, income source, and musculoskeletal conditions as direct parents of life satisfaction is more useful for policy than a model that predicts life satisfaction with $R^2 = 0.422$ by exploiting 58 correlated

features including proxy health variables, even if the latter appears superior by standard performance metrics.

Shmueli (2010) argues that the choice between predictive and explanatory modelling should be made explicitly and in advance, as the two goals require different model selection criteria, different validation strategies, and different interpretations of results. This study demonstrates that the choice also has different consequences for what is learned: the ML pipeline optimised for cross-validated R^2 identifies proxy-dominated feature rankings that are statistically accurate but causally ambiguous, while the BN pipeline optimised for structural plausibility recovers parsimonious causal hypotheses that are less predictively powerful but more directly actionable.

6.3. Data quality findings

The WHR Synthetic null result and the proxy dominance findings on Student Depression and the Canadian Survey collectively constitute a set of data quality lessons that extend beyond the specific datasets involved.

The synthetic dataset case study demonstrates that data presented on public platforms under the label of real-world survey data can contain no genuine predictive signal. The unanimous null result across all model families - including the BN's failure to learn any edges - provides the most rigorous possible confirmation of this absence. Importantly, this diagnosis was available at the baseline modelling stage, before any interpretability or causal analysis was attempted. The lesson is that baseline signal verification should be a standard first step in any comparative modelling pipeline applied to public datasets of uncertain provenance.

The proxy dominance findings raise a different data quality concern. The suicidal ideation variable in the Student Depression dataset is not a data error; it is a valid, reliably measured item. The problem is interpretive: a model that achieves high F1 by detecting suicidal ideation is not learning to predict depression onset from antecedent risk factors but is effectively detecting a correlated symptom.

The Canadian Survey proxy variables - `Mental_health_state` and `Gen_health_state` - present a related but distinct issue. These are self-rated health items collected in the same survey wave as the life satisfaction target, and they likely reflect the same underlying evaluative state from different angles. A model that relies on these variables is essentially predicting life satisfaction from life satisfaction, which inflates R^2 without providing causal insight. The sensitivity analysis

demonstrates that genuine predictive information remains after their removal, but the headline R^2 of 0.422 gives an overly optimistic impression of how much is understood about the determinants of individual life satisfaction.

These findings suggest that wellbeing datasets routinely contain features that are co-measurements or lagged reflections of the outcome rather than genuine upstream predictors, and that standard importance metrics including SHAP will elevate these features to top positions without flagging their problematic status. Domain knowledge and sensitivity analysis are both necessary to identify and quantify proxy influence - neither is standard in applied machine learning pipelines, and both should be.

6.4. Limitations

Several limitations of this study should be acknowledged, organised by their relevance to specific analytical choices.

Observational data and causal inference. All five datasets are observational, and none supports causal identification without additional assumptions. The causal language used in discussing BN structures - parents, direct causes, pathways - is shorthand for conditional independence claims consistent with a causal interpretation under the assumption that the learned DAG approximates the true data-generating process. The DAGs identified here are better described as causally plausible hypotheses consistent with domain knowledge and the observed correlation structure, rather than confirmed causal facts. Genuine causal identification would require experimental variation, longitudinal data, or instrumental variable designs, none of which are available here.

Cross-sectional measurement. Four of the five datasets are cross-sectional, measuring all variables at a single point in time. This limits the BN's ability to establish temporal precedence, which is a necessary though not sufficient condition for causal inference. The forbidden edges lists encoding temporal constraints are based on theoretical ordering rather than observed timing, and this ordering could be challenged for some variable pairs.

BN evaluation asymmetry. BN predictive performance is reported on the full dataset while ML performance is cross-validated. This asymmetry means that direct numerical comparisons overstate BN performance relative to a true out-of-sample evaluation. The magnitude is likely modest for the large datasets (Student Depression $n = 27,870$; Canadian Survey $n \approx 104,000$) and potentially more consequential for MH Tech ($n = 995$).

Hyperparameter optimisation. ML models are evaluated at default hyperparameter settings rather than being systematically tuned. This choice is deliberate - the goal is comparison across model families rather than maximum performance - but it means reported figures represent conservative estimates of what each model family can achieve. The practical effect is likely largest for XGBoost and smallest for logistic and linear regression.

Dataset representativeness. The five datasets vary considerably in representativeness. The OSMI Mental Health in Tech survey is a convenience sample with self-selection bias that limits generalisation. The Kaggle Student Depression dataset has an unclear sampling frame. The Canadian Community Health Survey is a nationally representative probability sample. The WHR Gallup data uses stratified random sampling within countries and is the most credible source for cross-national comparisons.

Discretisation for BN inference. Continuous variables were quantile-discretised into five bins for BN estimation and inference. Discretisation inevitably loses information, and the choice of five bins is a pragmatic compromise between granularity and CPT sparsity. Different discretisation choices could produce different learned structures, and a systematic sensitivity analysis over discretisation granularity was not conducted.

Chapter 7 - Conclusion

7.1. Summary of findings

This thesis compared statistical, machine learning, hybrid, and Bayesian Network models for predicting and explaining psychological wellbeing across five datasets spanning individual and societal levels of measurement. Five findings emerge consistently from the comparative design.

First, predictive performance is substantially higher at the societal level than the individual level, independent of model choice - XGBoost achieves $CV R^2 = 0.850$ on WHR Gallup versus 0.422 on the Canadian Survey, a gap reflecting the noise-suppressing effect of aggregation rather than any modelling artefact.

Second, hybrid models provide negligible predictive gains across all datasets (range: -0.010 to $+0.018$), indicating no stable middle ground between ensemble accuracy and linear interpretability.

Third, BN implausibility rates vary predictably with data quality - from 11% on WHR Gallup to 79% on Student Depression - but domain constraints consistently recover theoretically coherent parent sets at a modest, quantifiable BIC cost across all four substantive datasets.

Fourth, the variables identified by restricted BNs as direct parents of the target converge with post-proxy-exclusion SHAP rankings across three of four datasets, providing mutually reinforcing cross-method evidence for the identified predictors.

Fifth, the cross-validated pipeline correctly returns near-zero performance on the WHR Synthetic null dataset across all model families, demonstrating that baseline signal verification is a reliable and necessary first step in applied comparative modelling.

7.2. Contributions

This thesis makes contributions at three levels: methodological, empirical, and practical.

Methodological contributions. The primary contribution is the systematic application of a four-family comparative pipeline to wellbeing data at both individual and societal levels. The five-dataset design enables generalisations not possible from single-dataset comparisons - that hybrid models add negligible value, that BN implausibility rates vary predictably with data quality, and that the macro-versus-micro performance gap is model-independent are all conclusions requiring multi-dataset evidence.

A second contribution is the formalisation of the BIC cost of constraints as a diagnostic metric, providing a dataset-agnostic quantification of the tension between statistical fit and domain plausibility transferable to any BN estimation exercise.

A third contribution is the demonstration that proxy variable identification and sensitivity analysis should be standard pipeline components. The sensitivity protocol applied here - re-running the full pipeline after removing candidate proxies and comparing importance rankings with BN parent sets - provides a replicable procedure for this check.

Empirical contributions. This thesis provides a cross-validated, multi-method characterisation of wellbeing predictors across four distinct populations. The consistent identification of food security and income source as direct parents of individual life satisfaction reinforces the OECD framework's emphasis on basic needs (OECD, 2020). Academic pressure and financial stress are confirmed as the primary modifiable antecedents of student depression by both BN and post-proxy SHAP analysis. GDP per capita is confirmed as the sole direct parent of national life satisfaction, with health and generosity as upstream mediators, extending the World Happiness Report framework (Helliwell et al., 2023).

Practical contributions. The practical contribution is a set of empirically grounded model selection guidelines: use ensemble methods for predictive accuracy; use linear models when interpretable coefficients are required; do not expect hybrid pipelines to outperform the best single model; use BNs with domain constraints for parsimonious causal structures; always verify baseline signal quality; and always conduct proxy sensitivity analysis before interpreting SHAP rankings as causal or policy-relevant.

7.3. Future work

Several directions follow naturally from the findings and limitations of this thesis.

Longitudinal data. Extending the BN pipeline to longitudinal survey data, where temporal ordering can be directly incorporated into the forbidden edges structure, would substantially strengthen the causal claims made here. Dynamic Bayesian Networks are a natural vehicle for this extension.

Causal discovery beyond BNs. Score-based methods such as GES and constraint-based methods such as the PC algorithm make different assumptions and have different sensitivity profiles. A

systematic comparison on the same datasets would clarify whether the structural findings here are robust across the causal discovery landscape.

Intervention modelling. The BN framework supports intervention modelling through Pearl's do-calculus, allowing queries about the expected change in a target distribution if a variable were intervened upon rather than conditioned on (Pearl, 2009). This would move the analysis from causal description to policy simulation.

Fairness and subgroup analysis. This thesis treats each dataset as a single population. In the MH Tech and Student Depression datasets, assessing whether predictive performance and causal structures are consistent across demographic subgroups would have direct implications for any clinical or occupational health deployment.

This thesis has demonstrated that the choice of modelling approach in wellbeing research is not merely a technical decision but a substantive one, with direct consequences for what is learned, what can be communicated, and what can be acted upon. The four model families examined here are not competing answers to the same question; they are tools suited to different questions, and their appropriate use depends on a clear articulation of whether the goal is prediction, explanation, or causal identification. The consistent application of all four families to the same datasets has made the costs and benefits of each choice empirically visible rather than theoretically assumed, and it is hoped that the guidelines and diagnostic tools developed here will be of use to researchers navigating this choice in their own work.

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